

3 TIMED TESTS · 50 QUESTIONS EACH

Gastrointestinal

Practice tests of USMLE-style questions drawn from the website quizzes, with answers and explanations.

For educational use only. This is a study aid, not a clinical reference.

Instructions

60 minutes
for each test

50 questions
in each test

About 72 seconds
for each question

Every question comes from the website. These tests are built from the quizzes on the Clinical Conditions website, 150 of its 532 questions, so if you have done the website quizzes you will meet questions you have seen before. For the truest measure, take a test before using the website quiz for these topics, and treat a remembered answer as revision rather than a sign of mastery.

TAKING A TEST

- Take each test in one sitting, with a timer set to 60 minutes. Do not pause the timer.
- Each question has five options and one best answer. Choose the single best option, even if more than one seems reasonable.
- Fill in the answer sheet after the last question of each test, or use a separate piece of paper.
- Do not use notes, the website, or other references. The abbreviation list on the next page is allowed.
- There is no penalty for guessing, so answer every question. An unanswered question counts as incorrect.
- When time ends, stop, even if you have not finished.

ABOUT THE TESTS

- No question appears in more than one test, so the three tests can be taken in any order.
- Every test has 15 physiology and 35 clinical questions, covering all 20 physiology topics and all 32 clinical topics, with a similar mix of Step 1 questions (mechanisms, microbiology, and pathology) and Step 2 questions (diagnosis, next step, and management).
- Questions are in random order, and the topic being tested is not shown until the answers.

SCORING AND REVIEW

- Your score is the number of correct answers. The answer key at the start of each answer section has the same layout as the answer sheet, so you can mark one against the other.
- Then read the explanations. Each one names the topic and step, so you can see what to revisit.
- Answers follow the course material. Where a guideline differs, the explanation says so under "Additional information".

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Abbreviations

ALLOWED DURING A TEST

6-MP 6-Mercaptopurine

ABX Antibiotics

ALP Alkaline phosphatase

ALT Alanine aminotransferase

AMA Anti-mitochondrial antibody

anti-CCP Anti-cyclic citrullinated peptide antibody

anti-dsDNA Anti-double-stranded DNA antibody

anti-HBc Antibody to hepatitis B core antigen

anti-HDV Antibody to hepatitis D virus

anti-tTG Anti-tissue transglutaminase antibody

AST Aspartate aminotransferase

ATP7A Copper-transporting ATPase A

ATP7B Copper-transporting ATPase B

BUN Blood urea nitrogen

cAMP Cyclic adenosine monophosphate

cAMP Cyclic adenosine monophosphate

CBD Common bile duct

CD117 The KIT receptor tyrosine kinase, a GIST marker

CD4 Cluster of differentiation 4

CDH1 The gene for E-cadherin, a cell adhesion protein

CEA Carcinoembryonic antigen

CK Creatine kinase

CN Cranial nerve

CN X Cranial nerve X (the vagus nerve)

COPD Chronic obstructive pulmonary disease

CRP C-reactive protein

CSF Cerebrospinal fluid

CT Computed tomography

CT AP Computed tomography of the abdomen and pelvis

CTR1 Copper transporter 1

CXR Chest x-ray

CYP Cytochrome P450

CYP2C19 Cytochrome P450 2C19

CYP2C9 Cytochrome P450 2C9

CYP2E1 Cytochrome P450 2E1

CYP3A4 Cytochrome P450 3A4

CYP7A1 Cholesterol 7α-hydroxylase; Cholesterol 7α-hydroxylase, the enzyme that makes bile salts from cholesterol

DMT1 Divalent metal transporter 1

EGD Esophagogastroduodenoscopy

ERCP Endoscopic retrograde cholangiopancreatography

EUS Endoscopic ultrasound

FAST Focused assessment with sonography in trauma

FBC Full blood count

FODMAP Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols

FODMAP Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols

G6PD Glucose-6-phosphate dehydrogenase

GE Gastroesophageal

GERD Gastroesophageal reflux disease

GGT Gamma-glutamyl transferase

GI Gastrointestinal

GLUT2 Glucose transporter 2

H1 Histamine receptor 1

H2 Histamine receptor 2

H2R Histamine H2 receptor

HBIG Hepatitis B immune globulin

HBsAg Hepatitis B surface antigen: current infection

HBV Hepatitis B virus

HCl Hydrochloric acid

HCV Hepatitis C virus

HDV Hepatitis D (delta) virus

HELLP Hemolysis, elevated liver enzymes, low platelets

HER2 Human epidermal growth factor receptor 2

HFE Hereditary hemochromatosis gene

HIDA Hepatobiliary iminodiacetic acid scan (cholescintigraphy)

HIV Human immunodeficiency virus

HIV Human immunodeficiency virus

HLA Human leukocyte antigen

HSV Herpes simplex virus

IBD Inflammatory bowel disease

IBS Irritable bowel syndrome

IBS-D Irritable bowel syndrome with diarrhea

ICU Intensive care unit

IF Intrinsic factor

IgA Immunoglobulin A

IgG Immunoglobulin G

IgG4 Immunoglobulin G4

IgM Immunoglobulin M

IM Intramuscular

IM Intramuscular

INR International normalized ratio

IV Intravenous

KIT A receptor tyrosine kinase, also called CD117

KUB Kidneys, ureters, and bladder

LDH Lactate dehydrogenase

LDL Low-density lipoprotein

LDL Low-density lipoprotein

LFTs Liver function tests

M3 Muscarinic acetylcholine receptor 3

MALT Mucosa-associated lymphoid tissue

MASLD Metabolic dysfunction-associated steatotic liver disease (formerly NAFLD)

MCV Mean corpuscular volume

MRCP Magnetic resonance cholangiopancreatography

MRI Magnetic resonance imaging

NAPQI N-acetyl-p-benzoquinone imine; N-acetyl-p-benzoquinone imine, the toxic metabolite of paracetamol

NGT Nasogastric tube

NS3/4A Nonstructural protein 3/4A

NS5A Nonstructural protein 5A

NS5B Nonstructural protein 5B

NSAID Nonsteroidal anti-inflammatory drug

NSAID Nonsteroidal anti-inflammatory drug

p-ANCA Perinuclear anti-neutrophil cytoplasmic antibody

PAS Periodic acid-Schiff

PBC Primary biliary cholangitis (formerly primary biliary cirrhosis)

PCR Polymerase chain reaction

PepT1 Peptide transporter 1

PET Positron emission tomography

PPI Proton pump inhibitor

PPIs Proton pump inhibitors

QT The interval from the start of the QRS complex to the end of the T wave

RLQ Right lower quadrant

RNA Ribonucleic acid

RUQ Right upper quadrant

SGLT1 Sodium-glucose cotransporter 1

SMA Smooth muscle actin; Superior mesenteric artery

TIPS Transjugular intrahepatic portosystemic shunt

TPMT Thiopurine methyltransferase

TPMT Thiopurine methyltransferase

TPN Total parenteral nutrition

UDCA Ursodeoxycholic acid

V1 Vasopressin receptor 1

WBC White blood cells

Test A

50 questions | 60 minutes | Start the timer

1 A 45-year-old man from rural Southeast Asia, who often eats raw freshwater fish, presents with painless jaundice and weight loss. Imaging shows a mass at the confluence of the right and left hepatic ducts.

Which organism most likely increased his risk for this tumor?

- (A) Helicobacter pylori
- (B) Entamoeba histolytica
- (C) Hepatitis A virus
- (D) Clonorchis sinensis
- (E) Schistosoma haematobium

2 A 57-year-old man with untreated hereditary hemochromatosis develops fever, hypotension, and hemorrhagic bullae on his legs 1 day after eating raw oysters.

Which organism is most likely responsible?

- (A) Staphylococcus aureus
- (B) Vibrio vulnificus
- (C) Vibrio cholerae
- (D) Bacillus cereus
- (E) Listeria monocytogenes

3 A 42-year-old woman with obesity has had three episodes of epigastric pain radiating to her back, each starting about an hour after a large fried meal and settling within 3 hours. She is pain-free today, afebrile, with a soft, nontender abdomen. LFTs and white cell count are normal. Ultrasound shows several mobile, shadowing gallstones and a normal gallbladder wall.

Which is the most appropriate management?

- (A) No treatment; reassure
- (B) Elective laparoscopic cholecystectomy
- (C) IV ABX and admission for emergency surgery
- (D) HIDA scan
- (E) ERCP with sphincterotomy

4 A 25-year-old woman eats table sugar. Enzymes on the brush border of her enterocytes digest it before it can be absorbed.

Which enzyme acts, and what does it form?

- (A) Sucrase, forming glucose and galactose
- (B) Lactase, forming glucose and galactose
- (C) Lactase, forming glucose alone
- (D) Sucrase, forming glucose and fructose
- (E) Maltase, forming glucose and fructose

5 A 57-year-old woman with PBC takes ursodeoxycholic acid and starts cholestyramine for pruritus.

How should her medications be timed?

- (A) UDCA 1 hour before or 4–6 hours after cholestyramine
- (B) Cholestyramine only on days without ursodeoxycholic acid
- (C) Both taken together with breakfast
- (D) Cholestyramine at bedtime, ursodeoxycholic acid straight after
- (E) Ursodeoxycholic acid within 15 minutes after cholestyramine

6 A 72-year-old woman with long-standing pulmonary hypertension and severe tricuspid regurgitation has jugular venous distention, bilateral leg edema, and a tender, enlarged liver. Pressing on the liver increases her jugular venous pulsation. At autopsy, the cut surface of the liver shows dark red centrilobular areas alternating with pale tan areas.

Which mechanism best explains the liver findings?

- (A) Raised central venous pressure transmitted to the hepatic veins
- (B) Portal vein thrombosis
- (C) Fatty infiltration of hepatocytes
- (D) Thrombosis of the hepatic veins blocking outflow from the sinusoids
- (E) Occlusion of the hepatic artery

7 A 27-year-old woman has 8 months of recurrent lower abdominal cramping that improves after bowel movements, with alternating loose and hard stools. She has no weight loss, bleeding, or nocturnal symptoms. FBC, CRP, fecal calprotectin, and celiac serology are normal.

Which is the most appropriate next step?

- (A) CT AP
- (B) Empiric metronidazole for small bowel overgrowth
- (C) Reassurance and a trial of a low-FODMAP diet
- (D) Start oral prednisone for presumed IBD
- (E) Colonoscopy

8 A 56-year-old man has 2 days of right upper quadrant pain, jaundice, dark urine, and itching. He is afebrile. Bilirubin is 5.8 mg/dL and ALP is 410 U/L, with mildly raised AST/ALT and normal lipase. Ultrasound shows gallstones and a common bile duct dilated to 10 mm.

Which is the most appropriate next step?

- (A) ERCP with stone extraction
- (B) Laparoscopic cholecystectomy alone
- (C) CT-guided liver biopsy
- (D) HIDA scan
- (E) Repeat LFTs in 2 weeks

9 A 26-year-old woman has had diarrhea for 2 days and asks for loperamide.

Which finding should stop loperamide being given?

- (A) Cramping without blood or fever
- (B) Bloody stools with a fever of 39 °C
- (C) Diarrhea starting hours after eating reheated rice
- (D) Watery stools after a cruise
- (E) A history of IBS with diarrhea

10 A pharmacology student asks why the area postrema can respond to toxins in the blood when most of the brainstem cannot.

Which feature of the area postrema explains this?

- (A) Its ependyma actively pumps toxins in from the CSF
- (B) It lies in the floor of the third ventricle, beside the hypothalamus
- (C) Its capillaries lack tight junctions, so it sits outside the blood-brain barrier
- (D) It is bathed in lymph drained from the abdominal organs
- (E) Its neurons send axons out through the vagus nerve to the gut

11 A 19-year-old man is tackled during a football match and develops right upper quadrant pain. He also complains of pain at the tip of his right shoulder.

The shoulder pain is referred through which nerve?

- (A) Vagus nerve
- (B) Intercostal nerve
- (C) Greater splanchnic nerve
- (D) Phrenic nerve
- (E) Long thoracic nerve

12 A 69-year-old man in the ICU with cardiogenic shock is receiving high-dose noradrenaline. He develops abdominal distention and rising lactate. CT angiography shows diffuse narrowing of the mesenteric arteries without an embolus or thrombus.

Which is the most likely diagnosis?

- (A) Non-occlusive mesenteric ischemia
- (B) Mesenteric venous thrombosis
- (C) Chronic mesenteric ischemia
- (D) SMA embolus
- (E) Volvulus

13 A 2-week-old infant suddenly vomits green fluid and becomes unwell. Imaging suggests abnormal positioning of the bowel with a twisted mesentery.

Which developmental step failed?

- (A) Closure of the abdominal wall
- (B) Fusion of the pancreatic ducts
- (C) Recanalization of the solid duodenum after week 8
- (D) Separation of trachea and esophagus
- (E) The 270° rotation of the midgut around the SMA

14 A 45-year-old man's brother was recently diagnosed with HFE hereditary hemochromatosis. He feels well and asks to be tested.

Which is the best initial test?

- (A) Echocardiography
- (B) MRI of the liver
- (C) Liver biopsy with Prussian blue stain
- (D) Serum ceruloplasmin
- (E) Transferrin saturation and serum ferritin

15 A 58-year-old man with celiac disease who has not followed a gluten-free diet presents with weight loss, abdominal pain, and a small bowel perforation. Pathology of the resected segment shows a T-cell malignancy.

Which is the most likely diagnosis?

- (A) Enteropathy-associated T-cell lymphoma
- (B) Colorectal adenocarcinoma
- (C) Carcinoid tumor
- (D) Extranodal marginal zone (MALT) lymphoma
- (E) Burkitt lymphoma

16 A 66-year-old man with rheumatoid arthritis needs long-term NSAID therapy and has a history of a gastric ulcer. He is started on a drug that is a prostaglandin E1 analog to protect the gastric mucosa.

This drug is contraindicated in which group of patients?

- (A) Patients with asthma
- (B) Pregnant patients
- (C) Patients with renal impairment
- (D) Patients taking warfarin
- (E) Patients with hypothyroidism

17 A 38-year-old man allergic to penicillin starts H. pylori treatment. After drinking wine at a wedding, he has flushing, throbbing headache, and vomiting.

Which drug most likely caused this reaction?

- (A) Metronidazole
- (B) Omeprazole
- (C) Tetracycline
- (D) Clarithromycin
- (E) Levofloxacin

18 A 49-year-old woman with PBC starts ursodeoxycholic acid.

How does this drug help her?

- (A) It blocks viral reverse transcriptase in the hepatocyte
- (B) It displaces toxic bile salts and increases bile flow
- (C) It binds bile salts in the gut and blocks their reuptake
- (D) It inhibits CYP7A1, stopping synthesis of new bile salts
- (E) It suppresses T cells that are attacking her bile ducts

19 A 39-year-old woman with severe IBS-D has not responded to loperamide or a low FODMAP diet. Alosetron is considered.

Which serious adverse effect limits its use?

- (A) Tardive dyskinesia
- (B) Pancreatitis
- (C) Tendon rupture
- (D) Hepatic adenoma
- (E) Ischemic colitis

20 A patient has a stone in the common bile duct and develops an infection of the obstructed bile.

Which combination of findings makes up Charcot triad?

- (A) Painless jaundice, palpable gallbladder, weight loss
- (B) Fever, right upper quadrant pain, jaundice
- (C) Right upper quadrant pain, Murphy sign, fever
- (D) Jaundice, pruritus, pale stools
- (E) Fever, hypotension, confusion

21 A 79-year-old woman has small bowel obstruction. CT shows air in the biliary tree, dilated small bowel, and a 3.5-cm gallstone in the distal ileum. She is resuscitated.

Which is the most appropriate surgical procedure?

(A) Emergency cholecystectomy and fistula repair as the first step
(B) Colonoscopic stone removal
(C) Nasogastric decompression alone
(D) Right hemicolectomy
(E) Enterolithotomy to remove the stone from the bowel

22 A 58-year-old man on warfarin after a valve replacement has odynophagia, and endoscopy shows white plaques of Candida esophagitis. He starts a 14-day oral course of the first-line drug.

Which change in his warfarin therapy is most likely to be needed?

(A) Stopping warfarin, because the drug causes thrombocytopenia
(B) A higher warfarin dose, because the drug induces CYP enzymes
(C) A heparin bridge, because the drug blocks vitamin K absorption
(D) Closer INR checks, because the drug inhibits CYP2C9
(E) No change, because the drug acts only in the esophagus

23 A 30-year-old man has bloody diarrhea with tenesmus. Colonoscopy shows continuous inflammation from the rectum to the splenic flexure. His alkaline phosphatase is elevated, and MRCP shows multifocal strictures and dilations ("beading") of the intra- and extrahepatic bile ducts.

Besides colorectal cancer, this patient is at greatest risk for which malignancy?

(A) Gastric MALT lymphoma
(B) Small-bowel lymphoma
(C) Cholangiocarcinoma
(D) Hepatocellular carcinoma
(E) Pancreatic adenocarcinoma

24 A 62-year-old obese man with 20 years of poorly controlled GERD undergoes endoscopy. Biopsies from salmon-colored mucosa above the GE junction show columnar epithelium with goblet cells.

This patient is at greatest risk of developing which of the following?

(A) Squamous cell carcinoma of the upper esophagus
(B) Adenocarcinoma of the distal esophagus
(C) Gastric MALT lymphoma
(D) Zenker diverticulum
(E) Esophageal leiomyoma

25 A 22-year-old man has 18 hours of abdominal pain that began around the umbilicus and moved to the right lower quadrant, with anorexia and a low-grade fever. Lying on his left side, he has pain when his right hip is passively extended.

This finding suggests the inflamed appendix lies in which position?

(A) Anterior, just under McBurney point
(B) Left lower quadrant
(C) Retrocecal, against the iliopsoas muscle
(D) Pelvic, against the obturator internus
(E) Subhepatic

26 A 64-year-old woman with autoimmune gastritis has megaloblastic anemia, low serum B12, and reduced vibration sense in her feet.

Which is the most appropriate treatment?

(A) Oral folic acid alone
(B) Oral iron
(C) Proton pump inhibitor
(D) Intramuscular vitamin B12
(E) H. pylori eradication therapy

27 A 30-year-old woman at 35 weeks of pregnancy has nausea, vomiting, and abdominal pain. Her ALT is 420 IU/L and glucose 2.6 mmol/L.

Which criteria are used for the most likely diagnosis?

(A) Swansea criteria
(B) Maddrey score
(C) Child-Pugh score
(D) Rome criteria
(E) Charcot triad

28 A 67-year-old man has epigastric pain, early satiety, and weight loss. A hard, fixed node is felt above his left clavicle.

Which cancer does this sign most suggest?

(A) Cholangiocarcinoma
(B) Metastatic gastric cancer
(C) Colorectal adenocarcinoma
(D) Hepatocellular carcinoma
(E) Carcinoid tumor

29 A 30-year-old man with ulcerative colitis in remission on sulfasalazine is investigated for infertility and has a low sperm count.

Which change is most likely to help?

(A) Adding prednisone for 8 weeks
(B) Stopping all treatment
(C) Switching to azathioprine
(D) Switching to mesalazine
(E) Adding folic acid to sulfasalazine

30 A 23-year-old woman with acute liver failure after a paracetamol overdose becomes sweaty and less responsive in the ICU. Her INR is 4.2.

Which should be checked immediately?

(A) Thyroid-stimulating hormone
(B) Blood glucose
(C) Serum lipase
(D) Ceruloplasmin
(E) Serum ferritin

31 A 50-year-old woman with a duodenal ulcer and confirmed H. pylori infection needs a drug that coats the ulcer base and also kills the organism.

Which drug is most appropriate?

(A) Aluminum hydroxide gel
(B) Sucralfate
(C) Famotidine
(D) Misoprostol
(E) Bismuth subcitrate

32 A 59-year-old woman being treated for diverticulitis has persistent fever and pain after 3 days of IV ABX. CT shows a 6-cm pericolic fluid collection with an enhancing rim and no free air.

Which is the most appropriate next step?

(A) Colonoscopy
(B) Continue IV ABX alone and repeat CT in 1 week
(C) Increase dietary fiber
(D) Emergency colectomy with end colostomy
(E) CT-guided percutaneous drainage plus ABX

33 A 17-year-old boy with Wilson disease started D-penicillamine 4 months ago. He now has a rash, proteinuria, and a loss of taste.

Which is the best replacement for initial treatment?

(A) Zinc alone
(B) Trientine
(C) Deferoxamine
(D) Prednisolone
(E) Ursodeoxycholic acid

34 A 22-year-old man has mild, fluctuating jaundice with conjugated hyperbilirubinemia and normal enzymes. A liver biopsy done for another reason shows a black liver.

Which condition is most likely?

(A) Rotor syndrome
(B) Hemochromatosis
(C) Gilbert syndrome
(D) Crigler-Najjar syndrome
(E) Dubin-Johnson syndrome

35 A 78-year-old woman with a history of gallstones has 3 days of colicky abdominal pain, distention, vomiting, and no passage of stool or gas. CT shows air in the biliary tree, dilated small bowel loops, and a 3-cm calcified stone in the right lower quadrant.

Where is the stone most likely lodged?

(A) Ileocecal valve
(B) Duodenojejunal flexure
(C) Pylorus
(D) Sigmoid colon
(E) Ampulla of Vater

36 A baby is born to a mother who is HBsAg positive. The baby is healthy, at term, and weighs 3.4 kg.

Which is the most appropriate management for the newborn?

- (A) HBIG only
- (B) Tenofovir for the newborn for 6 months
- (C) Hepatitis B vaccine at 2 months of age
- (D) Hepatitis B vaccine plus HBIG within 12 hours of birth
- (E) Test the newborn for HBsAg at birth and treat only if positive

37 A 5-year-old boy had bloody diarrhea for 4 days after eating an undercooked hamburger. He is now pale with petechiae and decreased urine output. Bloods show hemoglobin 7.8 g/dL, platelets 42000/µL, creatinine 2.1 mg/dL and schistocytes on the blood smear.

Which intervention should be avoided?

- (A) Monitoring electrolytes and urine output
- (B) Careful IV fluid management
- (C) ABX therapy
- (D) Red blood cell transfusion if symptomatic
- (E) Dialysis if indicated

38 Between meals, a wave of regular, propulsive contractions sweeps through the small intestine about every 90–120 minutes.

What is this pattern?

- (A) The gastrocolic reflex triggered by gastric stretch
- (B) Haustral segmentation
- (C) The migrating motor complex
- (D) Receptive relaxation
- (E) Mass movement

39 A 52-year-old man is admitted with acute pancreatitis. Heart rate is 112/min, BUN is rising, and urine output is low.

Which is the most important initial treatment?

- (A) Total parenteral nutrition
- (B) Prophylactic IV carbapenem to prevent infected necrosis
- (C) Urgent contrast CT
- (D) Goal-directed IV fluids such as Hartmann solution
- (E) Emergency necrosectomy

40 A 28-year-old woman who has taken oral contraceptives for 10 years presents with sudden right upper quadrant pain and hypotension. CT shows a 7-cm liver mass with surrounding hemorrhage.

Which is the most likely underlying lesion?

- (A) Angiosarcoma
- (B) Focal nodular hyperplasia
- (C) Hemangioma
- (D) Hepatic adenoma
- (E) Hepatocellular carcinoma

41 A 71-year-old man who smokes has 3 months of progressive dysphagia to solids and liquids and has lost 11 kg. Barium swallow shows a "bird's beak" narrowing at the gastroesophageal junction.

Which is the most appropriate next step?

- (A) Trial of a calcium channel blocker
- (B) Botulinum toxin injection
- (C) Pneumatic dilation
- (D) Esophageal manometry
- (E) Upper endoscopy

42 Two hours after eating leftover fried rice from a takeaway shop, a 25-year-old man develops nausea and repeated vomiting. His symptoms resolve within 24 hours.

Which organism is most likely responsible?

- (A) Staphylococcus aureus
- (B) Salmonella enteritidis
- (C) Bacillus cereus
- (D) Enterotoxigenic E. coli
- (E) Clostridium perfringens

43 A 47-year-old man who drinks a bottle of vodka daily has fever, jaundice, and tender hepatomegaly after a week of heavy drinking. AST is 240 U/L, ALT is 95 U/L, GGT is elevated, and MCV is 104 fL.

Which is the most likely diagnosis?

- (A) Alcoholic hepatitis
- (B) Paracetamol toxicity
- (C) Ischemic hepatitis
- (D) Acute viral hepatitis A
- (E) MASLD

44 A 26-year-old woman's mother and aunt died of diffuse gastric cancer in their 40s. Genetic testing shows she carries a germline CDH1 mutation. Her EGD is normal.

Which management is most appropriate?

(A) H. pylori testing and eradication alone
(B) Subtotal distal gastrectomy
(C) Prophylactic total gastrectomy
(D) Annual CEA levels
(E) Endoscopy only if symptoms develop

45 A 61-year-old man using an inhaled corticosteroid for COPD and recently treated with broad-spectrum ABX has painful swallowing. Endoscopy shows adherent white plaques along the esophagus.

Which finding would most likely be seen on a biopsy of the plaques?

(A) Owl's eye inclusions in endothelial cells
(B) More than 15 eosinophils per high-power field
(C) Pseudohyphae and budding yeast
(D) Goblet cells in columnar epithelium
(E) Multinucleated giant cells

46 A 49-year-old woman with uncomplicated diverticulitis is managed as an outpatient with a liquid diet.

Which analgesic is preferred?

(A) Aspirin
(B) Ibuprofen
(C) Paracetamol
(D) Morphine
(E) Celecoxib

47 A 72-year-old man develops profuse watery diarrhea a week after a course of an antibiotic for a dental abscess.

Which antibiotic is a classic cause of this complication?

(A) Clindamycin
(B) Oral vancomycin
(C) Rifaximin
(D) Nitrofurantoin
(E) Fidaxomicin

48 A 31-year-old woman at 36 weeks' gestation has nausea, abdominal pain, and jaundice. Blood pressure is 128/80 mm Hg. Two possible diagnoses are being considered: acute fatty liver of pregnancy and HELLP syndrome.

Which finding most favors acute fatty liver of pregnancy?

(A) Platelet count of 60000/µL with schistocytes
(B) Raised LDH
(C) Proteinuria
(D) Blood pressure over 160/110 mm Hg
(E) Hypoglycemia with prolonged INR

49 A 59-year-old woman with epigastric pain and melena has been resuscitated, and her hemoglobin is 8.2 g/dL. She has no liver disease.

What is the most appropriate first investigation?

(A) Colonoscopy after bowel preparation
(B) Barium swallow
(C) Capsule endoscopy
(D) EGD within 24 hours
(E) CT angiography first

50 A 34-year-old man with ulcerative colitis has an elevated ALP of 480 U/L on routine bloods, with normal bilirubin. Ultrasound shows no gallstones. He is p-ANCA positive and AMA negative.

Which test is most appropriate to confirm the diagnosis?

(A) Serum IgG4 alone
(B) Liver biopsy
(C) HIDA scan
(D) Repeat ultrasound in 6 months
(E) MRCP

TEST A

Answer sheet

Name _____ Date _____ Score _____ / 50

1 (A) (B) (C) (D) (E)	11 (A) (B) (C) (D) (E)	21 (A) (B) (C) (D) (E)	31 (A) (B) (C) (D) (E)	41 (A) (B) (C) (D) (E)
2 (A) (B) (C) (D) (E)	12 (A) (B) (C) (D) (E)	22 (A) (B) (C) (D) (E)	32 (A) (B) (C) (D) (E)	42 (A) (B) (C) (D) (E)
3 (A) (B) (C) (D) (E)	13 (A) (B) (C) (D) (E)	23 (A) (B) (C) (D) (E)	33 (A) (B) (C) (D) (E)	43 (A) (B) (C) (D) (E)
4 (A) (B) (C) (D) (E)	14 (A) (B) (C) (D) (E)	24 (A) (B) (C) (D) (E)	34 (A) (B) (C) (D) (E)	44 (A) (B) (C) (D) (E)
5 (A) (B) (C) (D) (E)	15 (A) (B) (C) (D) (E)	25 (A) (B) (C) (D) (E)	35 (A) (B) (C) (D) (E)	45 (A) (B) (C) (D) (E)
6 (A) (B) (C) (D) (E)	16 (A) (B) (C) (D) (E)	26 (A) (B) (C) (D) (E)	36 (A) (B) (C) (D) (E)	46 (A) (B) (C) (D) (E)
7 (A) (B) (C) (D) (E)	17 (A) (B) (C) (D) (E)	27 (A) (B) (C) (D) (E)	37 (A) (B) (C) (D) (E)	47 (A) (B) (C) (D) (E)
8 (A) (B) (C) (D) (E)	18 (A) (B) (C) (D) (E)	28 (A) (B) (C) (D) (E)	38 (A) (B) (C) (D) (E)	48 (A) (B) (C) (D) (E)
9 (A) (B) (C) (D) (E)	19 (A) (B) (C) (D) (E)	29 (A) (B) (C) (D) (E)	39 (A) (B) (C) (D) (E)	49 (A) (B) (C) (D) (E)
10 (A) (B) (C) (D) (E)	20 (A) (B) (C) (D) (E)	30 (A) (B) (C) (D) (E)	40 (A) (B) (C) (D) (E)	50 (A) (B) (C) (D) (E)

Fill one bubble for each question. The answer key at the start of the Test A answers uses the same layout.

Test B

50 questions | 60 minutes | Start the timer

1 A 60-year-old man with cirrhotic ascites is treated with sodium restriction and two diuretics. Months later, he has tender breast enlargement and a potassium of 5.9 mmol/L.

Which drug is responsible?

(A) Spironolactone
(B) Lactulose
(C) Propranolol
(D) Furosemide
(E) Terlipressin

2 A 55-year-old man receiving cisplatin is given aprepitant, ondansetron, and dexamethasone. The dexamethasone dose is lowered.

Why is the dexamethasone dose lowered?

(A) Aprepitant displaces dexamethasone from albumin
(B) Aprepitant inhibits CYP3A4, raising dexamethasone levels
(C) Dexamethasone induces CYP3A4, so aprepitant levels fall too low
(D) Aprepitant adds glucocorticoid activity of its own
(E) Ondansetron blocks the renal excretion of dexamethasone

3 A 56-year-old woman with primary biliary cholangitis has fingers that turn white in the cold, tight skin over her fingers, small dilated blood vessels on her face, and difficulty swallowing.

Which antibody is most likely positive for her second condition?

(A) Anti-Jo-1 antibody
(B) Anti-CCP antibody
(C) Anti-centromere antibody
(D) Anti-Scl-70 (topoisomerase I) antibody
(E) Anti-dsDNA antibody

4 A student traces cholecalciferol from the skin to its active form.

What is the correct order?

(A) Kidney, then liver, then skin
(B) Liver, then skin, then kidney
(C) Skin, then kidney, then liver
(D) Skin, then liver, then kidney
(E) Liver, then kidney, then skin

5 A 38-year-old woman with poorly controlled type 2 diabetes develops severe epigastric pain radiating to her back. Her blood sample looks milky, and lipase is markedly raised. She does not drink alcohol, and ultrasound shows no gallstones.

Which laboratory finding most likely explains her pancreatitis?

(A) Elevated serum ferritin
(B) Serum triglycerides of 2400 mg/dL
(C) Positive anti-mitochondrial antibody
(D) Serum calcium of 8.0 mg/dL
(E) Elevated serum IgG4

6 A 60-year-old man with alcohol-related cirrhosis has moderate ascites. He has stopped drinking. Creatinine and sodium are normal.

Which is the most appropriate initial treatment?

(A) Fluid restriction to 500 mL per day, with sodium intake left unrestricted
(B) Furosemide alone
(C) Immediate TIPS
(D) Weekly large-volume paracentesis
(E) Sodium restriction and spironolactone, with furosemide added as needed

7 A 4-week-old first-born boy has had projectile, nonbilious vomiting after every feed for 3 days and is hungry afterward. A firm mass is felt in the right upper quadrant.

Which metabolic picture is expected?

(A) Hyperchloremic metabolic acidosis
(B) Hyperkalemic metabolic acidosis
(C) Hypochloremic, hypokalemic metabolic alkalosis
(D) High anion gap metabolic acidosis
(E) Respiratory acidosis

8 A 48-year-old man is diagnosed with HFE hereditary hemochromatosis. His hemoglobin is 15.2 g/dL, ferritin is 1600 ng/mL, and he has no cardiac disease.

Which is the most appropriate treatment?

(A) Low-iron diet alone
(B) Liver transplant
(C) Therapeutic phlebotomy
(D) D-penicillamine
(E) Oral deferasirox

9 A 24-year-old man has had recurrent sinusitis and two episodes of giardiasis. Serum IgA is undetectable, with normal IgG and IgM. He is booked for elective surgery.

Which precaution is most important if he needs a blood transfusion?

(A) Give IV immunoglobulin first
(B) Give only irradiated blood
(C) Avoid transfusion of platelets only
(D) Use blood products without IgA
(E) Premedicate with an antihistamine and use any blood

10 A 74-year-old woman with A-fib who is not anticoagulated has sudden, severe periumbilical pain that began 3 hours ago. She is writhing in pain, but her abdomen is soft with only mild tenderness. Serum lactate is elevated.

Which is the most appropriate diagnostic test?

(A) Colonoscopy with mucosal biopsy
(B) CT angiography of the abdomen
(C) Barium enema with air contrast
(D) Plain abdominal x-ray
(E) Abdominal ultrasound

11 A 58-year-old man has had untreated achalasia for 25 years, with a markedly dilated esophagus full of retained food.

He is at increased risk of which malignancy?

(A) Leiomyosarcoma of the esophageal muscle wall
(B) Squamous cell carcinoma of the esophagus
(C) Hepatocellular carcinoma
(D) Adenocarcinoma of the proximal stomach
(E) Gastric MALT lymphoma

12 A 78-year-old man has sudden, painless passage of a large volume of bright red blood per rectum. He is otherwise well, with a soft, nontender abdomen.

Which is the most likely cause?

(A) Anal fissure
(B) Diverticulosis
(C) Ischemic colitis
(D) Ulcerative colitis
(E) Internal hemorrhoids

13 A 28-year-old woman at 20 weeks' gestation has 1 day of abdominal pain, now worst in the right side of the abdomen, with nausea and a temperature of 38.0°C. WBC is 15000/µL.

Which is the most appropriate initial imaging?

(A) Graded-compression ultrasound, followed by MRI if inconclusive
(B) No imaging; observe for 48 hours and repeat the white cell count
(C) Barium enema
(D) CT AP with contrast
(E) Abdominal x-ray

14 A 61-year-old man has had heartburn for years. Over the past 2 months he has had difficulty swallowing solid food and has lost 6 kg. His hemoglobin is 10.8 g/dL with a low MCV.

Which is the most appropriate next step?

(A) 24-hour pH monitoring
(B) H. pylori stool antigen test
(C) Upper endoscopy with biopsy
(D) 8-week trial of a PPI
(E) Esophageal manometry

15 A 50-year-old woman has sudden severe epigastric pain radiating to her back with vomiting. Lipase is 9 times the upper limit of normal, ALT is 320 U/L, and ALP is raised. She drinks alcohol rarely. Ultrasound shows gallstones.

Where is the obstructing stone most likely located?

(A) Gallbladder fundus
(B) Hepatopancreatic ampulla (of Vater)
(C) Cystic duct
(D) Right hepatic duct at the porta hepatis
(E) Common hepatic duct near the hilum

16 A 35-year-old man is admitted with full-thickness burns over 45% of his body surface area. On hospital day 3, he has an episode of coffee-ground emesis.

Which mechanism best explains this complication?

(A) Urease-mediated mucosal injury
(B) Inhibition of prostaglandin synthesis
(C) Hypovolemia leading to gastric mucosal ischemia
(D) Increased vagal stimulation of parietal cells
(E) Gastrin-secreting tumor

17 An 81-year-old woman with long-standing gallstones has chronic watery diarrhea and weight loss. Imaging shows air in the biliary tree and an abnormal connection between the gallbladder and the hepatic flexure of the colon.

Which best explains her diarrhea?

(A) Lactose intolerance
(B) Pancreatic enzyme deficiency
(C) Bile salts entering the colon directly
(D) Bacterial overgrowth in the stomach
(E) Small bowel obstruction

18 A 71-year-old woman has fever of 39.2°C, right upper quadrant pain, and jaundice. Over several hours she becomes confused, and her blood pressure falls to 82/48 mm Hg despite IV fluids. Ultrasound shows a dilated common bile duct with a stone.

Which is the most important definitive step after starting IV ABX and resuscitation?

(A) HIDA scan
(B) Oral ABX and outpatient follow-up
(C) Ursodeoxycholic acid
(D) Urgent ERCP for biliary drainage
(E) Elective cholecystectomy in 6 weeks

19 A 34-year-old man with HIV who stopped antiretroviral therapy a year ago presents with 2 weeks of painful swallowing. His CD4 count is 62/μL. Examination shows white plaques on the buccal mucosa that scrape off, leaving an erythematous base. He is hemodynamically stable and can tolerate oral fluids.

Which of the following is the most appropriate next step in management?

(A) Empiric oral fluconazole
(B) Oral acyclovir
(C) IV ganciclovir
(D) High-dose oral omeprazole
(E) Upper endoscopy with biopsy before any treatment

20 A 67-year-old woman takes her weekly bisphosphonate with a sip of water just before lying down. Days later, she has severe pain on swallowing.

Where are pills most likely to lodge and cause this injury?

(A) The gastric antrum
(B) The upper esophageal sphincter
(C) The gastroesophageal junction only
(D) The mid-esophagus
(E) The duodenal bulb

21 A 26-year-old man has his third episode in 6 months of severe nausea, vomiting, and abdominal cramps. He spends hours in hot showers, which help. He has used cannabis daily for 5 years. Between episodes he is well.

Which of the following is the most effective long-term treatment?

- (A) Amitriptyline prophylaxis for migraine
- (B) Pyloric botulinum injection
- (C) A scopolamine patch before each episode
- (D) Daily ondansetron
- (E) Stopping cannabis

22 A 38-year-old woman has a 3-cm liver lesion found by chance on ultrasound. Contrast MRI shows peripheral nodular enhancement that gradually fills in toward the center. She has no symptoms or liver disease.

Which is the most appropriate management?

- (A) Percutaneous biopsy
- (B) Surgical resection
- (C) Transarterial chemoembolization
- (D) Stop any oral contraceptives and repeat imaging in 1 month
- (E) Reassurance; no biopsy or treatment needed

23 A 63-year-old woman who takes naproxen daily develops sudden, severe epigastric pain that spreads across the abdomen. She is tachycardic, and the abdomen is rigid with rebound tenderness. An upright chest x-ray shows air under the right hemidiaphragm.

In addition to IV fluids, IV PPI, and ABX, which is the most appropriate next step?

- (A) Barium swallow to locate the perforation
- (B) Urgent upper endoscopy
- (C) Nasogastric tube and serial abdominal examinations only
- (D) H. pylori testing and observation
- (E) Emergency surgical repair

24 A 38-year-old woman has early satiety and weight loss. EGD shows a thick, rigid stomach without a discrete mass, and the stomach does not distend with air. Biopsy shows scattered cells with mucin pushing the nucleus to one side.

Loss of which protein best explains this tumor's growth pattern?

- (A) E-cadherin
- (B) Cyclin D1
- (C) KIT (CD117)
- (D) APC
- (E) HER2

25 A 62-year-old man on simvastatin starts triple therapy for H. pylori. A week later, he has diffuse muscle pain and a raised CK.

Which drug in his regimen is most responsible?

- (A) Clarithromycin
- (B) Omeprazole
- (C) Amoxicillin
- (D) Bismuth subcitrate
- (E) Metronidazole

26 A liver biopsy from a woman with acute liver failure in the third trimester of pregnancy is reviewed.

Which histologic finding is most characteristic of acute fatty liver of pregnancy?

- (A) Macrovesicular steatosis with Mallory-Denk bodies
- (B) PAS-positive globules in periportal hepatocytes
- (C) Noncaseating granulomas
- (D) Periportal fibrin deposits and hemorrhage
- (E) Microvesicular steatosis

27 A 74-year-old male smoker collapses with severe back and left flank pain. His blood pressure is 78/40 mm Hg, and a pulsatile mass is felt above the umbilicus.

What is the most appropriate first investigation?

- (A) CT KUB for a ureteric stone
- (B) CT angiogram of the chest, abdomen, and pelvis
- (C) Upright chest x-ray
- (D) Bedside ultrasound
- (E) Serum lipase

28 A 58-year-old man with alcohol-related cirrhosis and ascites has fever, diffuse abdominal tenderness, and new confusion. Diagnostic paracentesis shows an ascitic neutrophil count of 520/μL.

Which is the most appropriate treatment?

(A) Large-volume paracentesis with albumin, without ABX
(B) Oral rifaximin alone
(C) Emergency laparotomy
(D) Increase the diuretic dose
(E) IV third-generation cephalosporin plus IV albumin

29 A 31-year-old man is brought in after a motorcycle collision. He has right upper quadrant tenderness and abrasions over the right lower ribs. Blood pressure is 128/76 mm Hg and heart rate 88/min, and they remain stable after 1 L of crystalloid.

Which is the most appropriate next step?

(A) Upper endoscopy
(B) Diagnostic peritoneal lavage
(C) CT abdomen with IV contrast
(D) Discharge with outpatient follow-up
(E) Emergency laparotomy

30 A 45-year-old woman has bilateral ovarian masses and ascites. Ovarian biopsy shows mucin-filled cells with nuclei pushed to the side.

Where is the primary tumor most likely to be?

(A) Stomach
(B) Ovary
(C) Endometrium
(D) Kidney
(E) Thyroid

31 A 27-year-old man has bloating, gas, and watery diarrhea within hours of drinking milk. His symptoms stop when he avoids dairy.

Which treatment most directly addresses the deficient step?

(A) A gluten-free diet
(B) IM B12 replacement
(C) Lactase replacement
(D) Oral vancomycin
(E) Pancreatic enzyme replacement

32 A 26-year-old woman at 14 weeks of pregnancy has periumbilical pain that has moved to the RLQ over 18 hours, with anorexia and a temperature of 38.0 °C.

Which imaging is most appropriate first?

(A) Abdominal x-ray
(B) Ultrasound
(C) CT AP with contrast
(D) Upright CXR
(E) Colonoscopy

33 A 48-year-old woman has 18 hours of constant right upper quadrant pain radiating to her right shoulder, with nausea. Temperature is 38.4°C. She stops breathing in when the right upper quadrant is palpated. WBC is 15800/μL; bilirubin, ALP, and lipase are normal. Ultrasound shows gallstones, a 5-mm gallbladder wall, and fluid around the gallbladder.

Which is the most appropriate definitive management?

(A) Oral ursodeoxycholic acid
(B) ERCP
(C) Outpatient follow-up in 6 weeks
(D) IV ABX and laparoscopic cholecystectomy during this admission
(E) Percutaneous cholecystostomy as first-line in all patients

34 A 34-year-old woman taking combined oral contraceptives, who has had three pregnancies, is found to have cholesterol gallstones.

How does estrogen promote cholesterol gallstone formation?

(A) It increases gallbladder contraction
(B) It causes hemolysis
(C) It increases cholesterol secretion into bile
(D) It increases bile salt synthesis
(E) It deconjugates bilirubin

35 A 31-year-old man has chronic diarrhea, bloating, and weight loss. He eats a regular diet that includes bread and pasta.

Which is the best initial test for the most likely diagnosis?

(A) Hydrogen breath test
(B) Upper endoscopy with duodenal biopsy
(C) Fecal elastase
(D) Stool ova and parasites on three separate samples
(E) IgA anti-tTG antibodies with total serum IgA

36 A 19-year-old woman is brought in 6 hours after taking 30 tablets of 500 mg paracetamol. She has mild nausea but is otherwise well. Her paracetamol level is above the treatment line on the nomogram.

The antidote she should receive works mainly by which mechanism?

(A) Enhancing glucuronidation of paracetamol
(B) Inhibiting CYP2E1
(C) Binding paracetamol in the gut
(D) Replenishing glutathione to neutralize NAPQI
(E) Increasing renal excretion of paracetamol

37 A 67-year-old man on daily naproxen presents with 2 days of black, tarry stools and lightheadedness. His BUN is 64 mg/dL and creatinine 1.1 mg/dL.

Why is the BUN:creatinine ratio raised?

(A) Muscle breakdown releasing creatinine
(B) Blood proteins digested and absorbed from the upper gut
(C) Naproxen blocking urea excretion in the renal collecting duct
(D) Liver failure reducing urea synthesis
(E) Acute tubular necrosis from hypotension

38 A 19-year-old woman presents 5 hours after taking a large overdose of paracetamol.

How does the antidote prevent liver injury?

(A) It speeds glucuronidation of paracetamol
(B) It increases renal excretion of paracetamol
(C) It binds paracetamol in the gut lumen before it is absorbed
(D) It blocks CYP2E1 from forming NAPQI
(E) It replenishes glutathione to detoxify NAPQI

39 A 26-year-old woman at 30 weeks' gestation develops fever, nausea, and jaundice 5 weeks after returning from a trip where she drank untreated water. Within days she becomes confused, with an INR of 2.8.

Which virus most likely caused her illness?

(A) Hepatitis D virus
(B) Hepatitis B virus
(C) Hepatitis A virus
(D) Hepatitis E virus
(E) Hepatitis C virus

40 A 70-year-old man with macrocytic anemia and numb feet is given folic acid alone. His hemoglobin improves, but his gait worsens.

What best explains this?

(A) The anemia was caused by iron deficiency
(B) Folic acid toxicity damages peripheral nerves
(C) Folic acid raises phenytoin levels
(D) Folic acid lowers B12 absorption in the ileum
(E) Untreated B12 deficiency progresses despite the better anemia

41 A 58-year-old man in the ICU is given an IV PPI for stress ulcer prophylaxis. A colleague suggests adding sucralfate.

Why is this combination a poor choice?

(A) Both drugs compete for the same pump
(B) Both raise the risk of torsades de pointes
(C) Sucralfate binds and inactivates the PPI
(D) Sucralfate is absorbed and blocks CYP2C19
(E) Sucralfate needs acid to form its paste

42 A 47-year-old man with Crohn disease, stable on azathioprine, develops gout and is started on allopurinol. A month later, he has pancytopenia.

Which mechanism best explains this?

(A) Allopurinol blocks xanthine oxidase, which breaks down 6-MP
(B) Allopurinol displaces azathioprine from albumin
(C) Allopurinol induces TPMT, raising the toxic thioguanine metabolites
(D) Allopurinol causes marrow suppression on its own
(E) Allopurinol inhibits CYP3A4, slowing azathioprine

43 A 74-year-old man after hip surgery has acute colonic pseudo-obstruction, with a cecum of 11 cm and no mechanical cause. IV neostigmine is planned.

Which precaution is most important?

- (A) Checking G6PD activity first
- (B) Cardiac monitoring with atropine at hand
- (C) Giving it with a PPI to prevent stress ulcers
- (D) Pretreating with an H1 antagonist
- (E) Checking TPMT activity first

44 A 29-year-old woman taking oral contraceptives is diagnosed with Budd-Chiari syndrome on Doppler ultrasound. She has mild ascites, a normal mental status, INR 1.3 and no evidence of cirrhosis.

Which is the most appropriate initial treatment?

- (A) Anticoagulation and stopping the oral contraceptive
- (B) Immediate liver transplantation
- (C) Observation with repeat ultrasound in 6 months
- (D) Transjugular intrahepatic portosystemic shunt (TIPS) before anticoagulation
- (E) Large-volume paracentesis alone

45 Several children at a childcare center develop fever, abdominal cramps, and small-volume bloody, mucoid stools. The outbreak spreads person to person despite no shared food source. Stool culture grows a non-motile, lactose-nonfermenting gram-negative rod that does not produce H_2S.

Which organism is most likely responsible?

- (A) Enterotoxigenic E. coli
- (B) Bacillus cereus
- (C) Shigella
- (D) Norovirus
- (E) Salmonella

46 A 30-year-old man develops acute hepatitis A. On day 5 of jaundice his INR is 1.6, but his albumin is normal at 41 g/L.

Why has his albumin not yet fallen?

- (A) Inflammation increases albumin production
- (B) Albumin has a half-life of 14–21 days
- (C) Albumin is stored in the spleen and released in illness
- (D) Acute hepatitis spares the albumin-making cells
- (E) Albumin is also made by the kidney

47 A 30-year-old man has intense sympathetic activation immediately after a road crash.

Which effect on the stomach is expected, and through which fibers?

- (A) Stimulation through efferent CN X fibers
- (B) Inhibition through efferent CN X fibers
- (C) Stimulation through sympathetic fibers
- (D) Inhibition through thoracolumbar sympathetic fibers
- (E) No effect, as only CN X fibers reach the stomach

48 A 29-year-old man with moderate-to-severe Crohn disease that has not responded to corticosteroids and azathioprine is to start infliximab.

Which test should be performed before starting therapy?

- (A) Serum ceruloplasmin
- (B) Thiopurine methyltransferase (TPMT) enzyme activity level
- (C) Pulmonary function tests
- (D) Interferon-gamma release assay (or tuberculin skin test)
- (E) Anti-Saccharomyces cerevisiae antibodies

49 A 33-year-old woman with diarrhea-predominant IBS and persistent abdominal pain is started on low-dose amitriptyline at night.

Which adverse effects is she most likely to experience?

- (A) Dry mouth and constipation
- (B) Diarrhea and hypersalivation
- (C) Hyperkalemia and gynecomastia
- (D) Bradycardia and miosis
- (E) Weight loss and insomnia

50 A 66-year-old man with painless jaundice has a CT showing a 2.5-cm mass in the head of the pancreas, with dilation of both the common bile duct and the pancreatic duct. There is no vascular invasion or metastasis.

Which is the most appropriate definitive treatment?

(A) Pancreaticoduodenectomy (Whipple procedure)
(B) Observation with repeat CT in 3 months
(C) Cholecystectomy
(D) Distal pancreatectomy
(E) ERCP stent alone as definitive treatment

Answer sheet

Name _____ Date _____ Score _____ / 50

1 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	11 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	21 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	31 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	41 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
2 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	12 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	22 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	32 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	42 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
3 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	13 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	23 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	33 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	43 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
4 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	14 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	24 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	34 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	44 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
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6 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	16 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	26 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	36 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	46 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
7 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	17 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	27 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	37 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	47 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
8 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	18 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	28 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	38 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	48 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
9 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	19 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	29 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	39 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	49 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
10 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	20 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	30 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	40 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	50 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Fill one bubble for each question. The answer key at the start of the Test B answers uses the same layout.

Test C

50 questions | 60 minutes | Start the timer

1 A 23-year-old woman who smokes has crampy right lower quadrant pain, non-bloody diarrhea, and weight loss. She also has a perianal fistula. Colonoscopy shows patchy ulcers in the terminal ileum and ascending colon separated by normal mucosa, and the rectum is spared.

Which biopsy finding is most likely?

(A) Noncaseating granulomas with transmural inflammation
(B) PAS-positive foamy macrophages
(C) Caseating granulomas
(D) Crypt abscesses limited to the mucosa
(E) Villous atrophy with intraepithelial lymphocytes

2 A 52-year-old woman has a positive anti-HCV antibody on routine one-time screening. She feels well, and her liver enzymes are normal.

Which is the most appropriate next step?

(A) HCV RNA by PCR
(B) Start direct-acting antiviral therapy
(C) Repeat anti-HCV antibody in 6 months
(D) Reassure her that she is immune
(E) Liver biopsy

3 A 19-year-old traveler has had high fever for 3 weeks. He suddenly develops severe abdominal pain, and CT shows free air near the distal ileum.

Which structure is most likely to have perforated?

(A) A duodenal ulcer
(B) A necrotic Peyer patch
(C) The appendix
(D) A bleeding, then perforated Meckel diverticulum
(E) A sigmoid diverticulum

4 A 46-year-old man has 12 hours of severe epigastric pain radiating to his back, with vomiting. He drinks heavily. Lipase is 5 times the upper limit of normal. He is hemodynamically stable.

Which is the most appropriate next step to confirm the diagnosis?

(A) No further test is needed
(B) Contrast CT abdomen immediately
(C) Repeat lipase in 24 hours
(D) Endoscopic ultrasound
(E) MRCP

5 A 28-year-old man wants fast relief of occasional heartburn after large meals.

Which drug class acts on HCl already in the lumen?

(A) H2R antagonists
(B) Antacids
(C) Misoprostol
(D) PPIs
(E) Sucralfate

6 A 62-year-old man who worked for decades in a plastics manufacturing plant presents with weight loss and hepatomegaly. Biopsy shows a malignant tumor of endothelial origin.

Which exposure most likely caused this tumor?

(A) Aflatoxin
(B) Aniline dyes
(C) Benzene
(D) Vinyl chloride
(E) Asbestos

7 A 61-year-old woman with cirrhosis and hepatic encephalopathy is started on lactulose.

How does lactulose lower ammonia absorption?

- (A) It blocks ammonia transport across enterocytes
- (B) Bacterial fermentation acidifies the colon, trapping NH4+
- (C) It increases conversion of ammonia to urea in periportal hepatocytes
- (D) It binds ammonia directly in the small intestine
- (E) It kills urease-producing bacteria in the colon

8 A 48-year-old woman is given ondansetron before her first cycle of chemotherapy. She takes citalopram and has a borderline long QT interval.

Which adverse effect of ondansetron is the main concern?

- (A) Hyperprolactinemia
- (B) Ischemic colitis
- (C) Respiratory depression
- (D) Tardive dyskinesia
- (E) QT prolongation

9 A 44-year-old man with heavy alcohol use has fever, jaundice, and tender hepatomegaly. Bilirubin is 14 mg/dL and his Maddrey discriminant function is 48. There is no infection or GI bleeding.

Which treatment is most appropriate in addition to abstinence and nutrition?

- (A) N-acetylcysteine alone
- (B) Ursodeoxycholic acid
- (C) Liver transplant immediately
- (D) Phlebotomy
- (E) Prednisolone

10 A 24-year-old woman is stabbed in the right upper abdomen. Her blood pressure is 78/40 mm Hg and heart rate 136/min despite 2 units of blood. A FAST exam shows free fluid in Morison pouch.

Which is the most appropriate next step?

- (A) Repeat FAST in 30 minutes
- (B) CT abdomen with IV contrast
- (C) Angioembolization
- (D) Emergency laparotomy
- (E) Serial abdominal examinations

11 A 44-year-old woman after a kidney transplant has HSV esophagitis and is treated with IV acyclovir.

Which measure best prevents the drug's main kidney toxicity?

- (A) Good hydration during treatment
- (B) Adding a loop diuretic
- (C) Giving it only by mouth
- (D) Alkalinizing the urine with bicarbonate
- (E) Giving it with an NSAID

12 A 56-year-old woman has jaundice, dark urine, and pale stools.

Which finding best explains her dark urine?

- (A) Stercobilin reaching the urine
- (B) Urobilin falling in the urine
- (C) Conjugated bilirubin reaching the urine
- (D) Unconjugated bilirubin reaching the urine
- (E) Biliverdin reaching the urine

13 A 75-year-old man with a common bile duct stone has fever, RUQ pain, and jaundice. Blood cultures are drawn before ABX are started.

Which organism is most likely to grow?

- (A) Escherichia coli
- (B) Staphylococcus aureus
- (C) Pseudomonas aeruginosa
- (D) Streptococcus pneumoniae
- (E) Candida albicans

14 A physiologist notes that toxins in the blood, gastric irritation, motion, and fear can all cause vomiting.

Where do these different inputs converge?

- (A) The vestibular nuclei
- (B) The dorsal motor nucleus of the vagus
- (C) The hypothalamus
- (D) The nucleus ambiguus
- (E) The nucleus tractus solitarius

15 A 52-year-old man with chronic pancreatitis has constant epigastric pain despite stopping alcohol and smoking. Paracetamol alone has not been enough.

Which drug is the best next step before opioids?

(A) Pregabalin
(B) Hyoscine butylbromide
(C) Pancreatic enzymes
(D) Prednisolone
(E) Octreotide

16 A 29-year-old man with diarrhea-predominant IBS has urgent loose stools on workdays despite a low-FODMAP diet. He has no alarm features, and his workup is normal.

Which drug is most appropriate for his diarrhea?

(A) Senna
(B) Polyethylene glycol
(C) Lactulose
(D) Metoclopramide
(E) Loperamide

17 A 30-year-old pregnant woman with mild heartburn is started on famotidine.

Famotidine reduces acid secretion by which mechanism?

(A) Activating prostaglandin E1 receptors
(B) Blocking H2 receptors, lowering cAMP in parietal cells
(C) Irreversibly blocking the proton pump in parietal cells
(D) Blocking muscarinic M3 receptors
(E) Binding bile salts

18 A 20-year-old woman has RLQ pain. When the examiner presses deeply on her left lower quadrant, she feels pain in the right lower quadrant.

What is this finding called?

(A) Obturator sign
(B) Rovsing sign
(C) Cullen sign
(D) Psoas sign
(E) Murphy sign

19 A 63-year-old man has painless jaundice. Imaging shows strictures of the bile ducts and a diffusely enlarged, "sausage-shaped" pancreas. Serum IgG4 is markedly elevated, and no mass is seen.

Which is the most appropriate treatment?

(A) Pancreaticoduodenectomy
(B) Liver transplant
(C) Ursodeoxycholic acid alone
(D) Corticosteroids
(E) Broad-spectrum ABX

20 A physiology lecture explains why the liver tolerates occlusion of the hepatic artery better than many other organs tolerate loss of their arterial supply.

Which vessel provides most of the liver's blood flow?

(A) Inferior vena cava
(B) Splenic artery
(C) Portal vein
(D) Hepatic artery proper
(E) Hepatic veins

21 A 26-year-old woman in her first pregnancy, at 35 weeks, has 3 days of nausea, vomiting, and malaise, and is now jaundiced and confused. Glucose is 2.4 mmol/L (43 mg/dL), INR is 2.1, AST is 480 U/L, platelets are 160000/µL and blood pressure is 134/86 mm Hg.

Which is the most appropriate definitive management?

(A) N-acetylcysteine and observation
(B) IV magnesium sulfate and expectant management
(C) Ursodeoxycholic acid
(D) Corticosteroids for fetal lung maturity, then deliver at 37 weeks
(E) Prompt delivery

22 A 3-day-old boy has not passed meconium and has a distended abdomen. A contrast enema suggests a narrow rectosigmoid with dilated bowel above.

Which test confirms the diagnosis?

(A) Abdominal ultrasound for a double bubble
(B) Sweat chloride test
(C) Rectal suction biopsy
(D) Stool calprotectin
(E) 4-hour gastric emptying scintigraphy

23 A 58-year-old man has biopsy-proven gastric adenocarcinoma. EUS shows the tumor reaching the muscularis propria with enlarged nodes, and CT of the chest, abdomen, and pelvis shows no distant metastases. Surgery is being planned.

Which step should come before chemotherapy and surgery?

(A) Endoscopic submucosal dissection
(B) Staging laparoscopy with peritoneal washings
(C) H. pylori eradication alone
(D) Serum CEA to confirm the diagnosis
(E) Repeat EGD in 3 months to watch for further growth

24 A 68-year-old man with small bowel obstruction from adhesions has been managed with IV fluids and an NGT. Over 6 hours his pain becomes constant and severe, his temperature is 38.6 °C, pulse 120, and lactate rises.

What is the most appropriate next step?

(A) Repeat x-ray in 24 hours
(B) Continue NGT decompression for 72 hours
(C) Urgent surgery
(D) Start oral laxatives
(E) Give IV neostigmine

25 An abdominal x-ray of a 68-year-old woman with chronic gallstones shows a calcified gallbladder wall outlining the gallbladder.

This finding is associated with an increased risk of which condition?

(A) Gallbladder carcinoma
(B) Hepatocellular carcinoma
(C) Acute pancreatitis
(D) Primary biliary cholangitis
(E) Hepatic angiosarcoma

26 A researcher follows labeled copper from the duodenal lumen across an enterocyte and into the blood.

Which pair of transporters does it use?

(A) DMT1, then ferroportin
(B) CTR1, then ATP7A
(C) SGLT1, then GLUT2
(D) PepT1, then ATP7B
(E) ATP7B, then CTR1

27 A 41-year-old woman with a positive H. pylori stool antigen test had anaphylaxis to penicillin as a child. Local clarithromycin resistance is low.

Which triple therapy regimen is most appropriate?

(A) PPI + clarithromycin + metronidazole
(B) PPI + amoxicillin + rifabutin
(C) PPI alone for 8 weeks
(D) PPI + amoxicillin + levofloxacin
(E) PPI + amoxicillin + clarithromycin

28 An 80-year-old man with untreated gallstones has 2 days of repeated vomiting and epigastric fullness. CT shows air in the biliary tree, a distended stomach, and a large gallstone impacted in the duodenal bulb.

Which is the most likely diagnosis?

(A) Gallstone pancreatitis
(B) Gallstone ileus at the ileocecal valve
(C) Acute cholangitis
(D) Bouveret syndrome
(E) Mirizzi syndrome

29 A 19-year-old man has a 6-month history of hand tremor, slurred speech, and new irritability. His AST and ALT are mildly elevated. Slit-lamp examination shows a golden-brown ring at the edge of each cornea. Serum ceruloplasmin is low.

A defect in which process best explains his condition?

(A) Hepatocyte copper excretion into bile
(B) Intestinal iron absorption
(C) Secretion of alpha-1 antitrypsin from hepatocytes
(D) Conjugation of bilirubin
(E) Hepatic uptake of LDL cholesterol

30 A 52-year-old woman with cirrhosis has a rising creatinine and low urine sodium despite albumin. Hepatorenal syndrome is diagnosed, and a vasoconstrictor is added.

How does this vasoconstrictor improve kidney function?

(A) Blocking aldosterone in the collecting duct
(B) Dilating renal arteries through dopamine receptors
(C) Constricting splanchnic vessels through V1 receptors
(D) Raising plasma oncotic pressure in the kidney
(E) Inhibiting renin release through β1 block in the kidney

31 An embolus blocks the SMA in an older woman.

Which embryologic region of the gut does this artery supply?

(A) Cloaca
(B) Hindgut
(C) Foregut
(D) Midgut
(E) Vitelline duct

32 A 19-year-old man with asthma and atopic dermatitis presents to the emergency department with a piece of steak stuck in his throat. He reports years of intermittent dysphagia to solids. Endoscopy shows concentric esophageal rings and linear furrows. Biopsy shows more than 15 eosinophils per high-power field.

Which of the following is the most likely diagnosis?

(A) Candida esophagitis
(B) Schatzki ring
(C) Achalasia
(D) Eosinophilic esophagitis
(E) Gastroesophageal reflux disease

33 A 58-year-old man with cirrhosis has breast enlargement, small testes, reddened palms, and many small, spider-like blood vessels on his chest.

Which mechanism best explains these findings?

(A) Decreased albumin synthesis by the failing liver
(B) Vitamin K deficiency from reduced bile salt secretion
(C) Increased estrogen from reduced hepatic clearance
(D) Increased ammonia from reduced hepatic clearance
(E) Portal hypertension

34 A 63-year-old woman with autoimmune gastritis has finished 2 weeks of IM vitamin B12. She dislikes injections and asks about tablets.

Why can high-dose oral B12 work despite her lack of intrinsic factor?

(A) B12 is absorbed in the stomach rather than the ileum
(B) Oral B12 stimulates the remaining parietal cells
(C) PPIs are given with it to raise absorption
(D) The tablets contain IF to replace hers
(E) A small fraction is absorbed passively without IF

35 A 47-year-old woman has RUQ pain with ALP 320 IU/L, GGT 410 IU/L, and bilirubin 45 µmol/L. Ultrasound shows gallstones and a dilated CBD, but no stone is seen. She has no fever.

Which test is most appropriate next?

(A) CT AP with contrast
(B) AMA
(C) Liver biopsy
(D) MRCP or endoscopic ultrasound
(E) HIDA scan for cystic duct obstruction

36 A 66-year-old man in the ICU after major burns, on TPN and vasopressors, develops fever and leukocytosis. Ultrasound shows a distended gallbladder with a thickened wall and fluid around it, but no gallstones. He is too unstable for surgery.

Which is the most appropriate management?

(A) Oral ursodeoxycholic acid
(B) ERCP with sphincterotomy
(C) IV ABX and percutaneous cholecystostomy
(D) Observation only
(E) Elective cholecystectomy in 6 weeks

37 A 62-year-old woman has gallstones on ultrasound and recent RUQ pain. ALP and GGT are mildly raised, bilirubin is 1.8 mg/dL, and the common bile duct measures 7 mm. No stone is seen in the duct. She has no fever.

Which is the most appropriate next step?

(A) Immediate ERCP
(B) Repeat LFTs in 6 months
(C) HIDA scan
(D) Cholecystectomy without further imaging
(E) MRCP or endoscopic ultrasound

38 Two patients have chronic H. pylori infection. One strain carries a gene encoding a protein that is injected into gastric epithelial cells by a type IV secretion system.

Strains with this virulence factor carry a higher risk of which outcome?

(A) Eosinophilic esophagitis
(B) Pancreatic insufficiency
(C) Celiac disease
(D) Achalasia
(E) Gastric adenocarcinoma

39 A 71-year-old woman has taken omeprazole daily for 9 years. She presents with muscle cramps, tremor, and a prolonged QT interval.

Which electrolyte abnormality is most likely?

(A) Hyperkalemia
(B) Hypermagnesemia
(C) Hypomagnesemia
(D) Hypercalcemia
(E) Hypernatremia

40 A 26-year-old woman has bloating, foul-smelling stools, iron-deficiency anemia, and an itchy vesicular rash on her elbows and knees. Duodenal biopsy shows villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes.

This condition is most strongly associated with which HLA types?

(A) HLA-DR5
(B) HLA-B27
(C) HLA-DR3 and DR4
(D) HLA-DQ2 and DQ8
(E) HLA-DR2

41 A 29-year-old woman is diagnosed with celiac disease on serology and duodenal biopsy. She asks how her condition will be monitored once she starts treatment.

Which is the best way to monitor her response to a gluten-free diet?

(A) Rising HLA-DQ2 expression
(B) Repeat D-xylose test every month
(C) Falling IgA anti-tTG levels
(D) Serial fecal elastase
(E) Annual colonoscopy

42 A 41-year-old man with known chronic hepatitis B, who injects drugs, has a sudden flare of hepatitis with jaundice. HBsAg is positive, IgM anti-HBc is negative, and IgG anti-HBc and anti-HDV are positive.

Which statement about his new infection is most accurate?

(A) It is an HDV coinfection and will likely resolve
(B) It is an HDV superinfection and is likely to become chronic
(C) It can be prevented by the hepatitis C vaccine
(D) It is reactivation of HBV alone
(E) It is acute hepatitis A

43 A 50-year-old man has several gallstones found incidentally on an abdominal ultrasound done for kidney stones. He has never had abdominal pain, and his LFTs are normal.

Which is the most appropriate management of his gallstones?

- (A) Repeat ultrasound and LFTs every 6 months
- (B) ERCP with sphincterotomy
- (C) Elective laparoscopic cholecystectomy
- (D) Ursodeoxycholic acid for 2 years
- (E) No treatment; return if symptoms develop

44 A 26-year-old woman develops watery diarrhea 3 days after arriving in Mexico. She has 4 loose stools a day, no fever and no blood in the stool, and is drinking well.

Which is the most appropriate management?

- (A) Stool ova and parasites, then empirical metronidazole
- (B) Oral rehydration, with loperamide if needed
- (C) IV ciprofloxacin and admission
- (D) Oral vancomycin
- (E) Urgent colonoscopy

45 A 55-year-old woman with chronic HCV has ascites and a Child-Pugh class C score. Her team is choosing a direct-acting antiviral regimen.

Which drug class should her regimen avoid?

- (A) Nucleotide analogs
- (B) NS3/4A protease inhibitors
- (C) Bile acid sequestrants
- (D) NS5A inhibitors
- (E) NS5B inhibitors

46 A 38-year-old woman has dysphagia to solids and liquids and occasional regurgitation. Barium swallow suggests achalasia, and upper endoscopy shows no mass at the gastroesophageal junction.

Which test is the gold standard for confirming the diagnosis?

- (A) Esophageal manometry
- (B) Endoscopic ultrasound
- (C) CT chest
- (D) 24-hour esophageal pH monitoring
- (E) Repeat barium swallow

47 A 66-year-old man is in the ICU on mechanical ventilation for 4 days after a severe head injury. His platelet count is 48000/µL.

Which intervention best reduces his risk of stress-related gastric bleeding?

- (A) No intervention is needed
- (B) H. pylori testing
- (C) Routine daily upper endoscopy
- (D) Prophylactic misoprostol
- (E) Prophylactic PPI or H2 blocker

48 A 22-year-old man is tackled during a rugby match and develops left upper quadrant pain. Twenty minutes later he also reports pain at the tip of his left shoulder, which he did not injure. His pulse is 118/min and blood pressure 96/60 mm Hg.

Which mechanism best explains his shoulder pain?

- (A) Radicular pain from a lower cervical nerve root
- (B) Direct injury to the acromioclavicular joint
- (C) Somatoparietal pain from the inflamed lateral abdominal wall
- (D) Referred pain from nerves sharing a spinal level
- (E) Visceral pain from stretch of a hollow organ wall

49 A 77-year-old man is in the ICU on vasopressors for septic shock. Two days later he develops crampy left-sided abdominal pain followed by bloody diarrhea.

Which region of the colon is most vulnerable to this injury?

- (A) Terminal ileum
- (B) Mid-transverse colon
- (C) Splenic flexure
- (D) Cecum and ascending colon
- (E) Distal rectum

50 A 68-year-old man with several prior episodes of sigmoid diverticulitis reports passing air bubbles in his urine and has had 3 urinary tract infections in 4 months.

Which is the most likely explanation for his symptoms?

(A) Urethral stricture
(B) Emphysematous cystitis
(C) Bladder exstrophy
(D) Colovesical fistula
(E) Benign prostatic hyperplasia

TEST C

Answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	11 A B C D E	21 A B C D E	31 A B C D E	41 A B C D E
2 A B C D E	12 A B C D E	22 A B C D E	32 A B C D E	42 A B C D E
3 A B C D E	13 A B C D E	23 A B C D E	33 A B C D E	43 A B C D E
4 A B C D E	14 A B C D E	24 A B C D E	34 A B C D E	44 A B C D E
5 A B C D E	15 A B C D E	25 A B C D E	35 A B C D E	45 A B C D E
6 A B C D E	16 A B C D E	26 A B C D E	36 A B C D E	46 A B C D E
7 A B C D E	17 A B C D E	27 A B C D E	37 A B C D E	47 A B C D E
8 A B C D E	18 A B C D E	28 A B C D E	38 A B C D E	48 A B C D E
9 A B C D E	19 A B C D E	29 A B C D E	39 A B C D E	49 A B C D E
10 A B C D E	20 A B C D E	30 A B C D E	40 A B C D E	50 A B C D E

Fill one bubble for each question. The answer key at the start of the Test C answers uses the same layout.

Test A answers

Answer key | Explanations

1	D	11	D	21	E	31	E	41	E
2	B	12	A	22	D	32	E	42	C
3	B	13	E	23	C	33	B	43	A
4	D	14	E	24	B	34	E	44	C
5	A	15	A	25	C	35	A	45	C
6	A	16	B	26	D	36	D	46	C
7	C	17	A	27	A	37	C	47	A
8	A	18	B	28	B	38	C	48	E
9	B	19	E	29	D	39	D	49	D
10	C	20	B	30	B	40	D	50	E

1 D Clonorchis sinensis

Biliary malignancies | Step 1

A perihilar mass at the hepatic duct confluence (Klatskin tumor) is a cholangiocarcinoma. Liver flukes such as Clonorchis sinensis, acquired from raw freshwater fish, cause chronic bile duct inflammation and raise the risk. PSC and choledochal cysts are other risk factors.

WHY NOT THE OTHERS

(E) Schistosoma haematobium is linked to squamous cell carcinoma of the bladder.

2 B Vibrio vulnificus

Metabolic liver injury | Step 1

Iron overload favors iron-loving (siderophilic) bacteria such as Vibrio vulnificus, Yersinia, and Listeria. Vibrio vulnificus from raw shellfish can cause fulminant sepsis with hemorrhagic bullae, so patients are told to avoid raw shellfish.

3 B Elective laparoscopic cholecystectomy

Cholelithiasis | Step 2

Recurrent biliary colic (transient cystic duct obstruction, normal LFTs, no fever) is an indication for elective laparoscopic cholecystectomy. Asymptomatic gallstones need no treatment, but symptomatic ones do, because attacks recur and complications such as cholecystitis can follow.

WHY NOT THE OTHERS

(A) Reassurance alone is for asymptomatic gallstones.

(E) ERCP removes common bile duct stones; her normal LFTs argue against choledocholithiasis.

4 D Sucrase, forming glucose and fructose

Physiology | Digestion and absorption | Step 1

Sucrase converts sucrose to glucose and fructose, lactase converts lactose to glucose and galactose, and maltase converts maltose to glucose.

5 A UDCA 1 hour before or 4–6 hours after cholestyramine

Physiology | Hepatitis and bile salts | Step 1

Cholestyramine is a non-absorbed resin that binds bile salts and other drugs, including UDCA, warfarin, digoxin, and levothyroxine, so they are given 1 hour before or 4–6 hours after it.

6 A Raised central venous pressure transmitted to the hepatic veins

Vascular liver injury | Step 1

Right heart failure raises central venous pressure, which backs up into the hepatic veins and sinusoids. Zone 3 congestion and atrophy produce the mottled "nutmeg liver". Chronic congestion can lead to cardiac cirrhosis.

WHY NOT THE OTHERS

(B) Portal vein thrombosis causes portal hypertension without centrilobular congestion.

(D) Budd-Chiari syndrome also causes a nutmeg pattern, but JVD and hepatojugular reflux point to a cardiac cause.

7 (C) Reassurance and a trial of a low-FODMAP diet

IBS · Step 2

IBS is a disorder of gut-brain interaction diagnosed clinically (Rome V criteria) once alarm features are absent and basic tests are normal. First-line management is reassurance, education, and a low-FODMAP diet, with symptom-directed drugs as needed.

WHY NOT THE OTHERS

(E) Colonoscopy is not needed without alarm features in a patient under 45.

8 (A) ERCP with stone extraction

Choledocholithiasis · Step 2

Bilirubin over 4 mg/dL with a dilated CBD gives a high likelihood of choledocholithiasis, so proceed directly to ERCP to remove the stone, followed by cholecystectomy during the same admission. MRCP or endoscopic ultrasound is used when the likelihood is intermediate.

WHY NOT THE OTHERS

(B) Cholecystectomy alone leaves the CBD stone in place.

9 (B) Bloody stools with a fever of 39 °C

Physiology · Motility · Step 1

Loperamide is avoided with bloody diarrhea, fever, C. difficile, HUS, toxic megacolon, acute ulcerative colitis, and in children under 2. It acts on μ-opioid receptors in the gut, and P-gp keeps it out of the brain.

10 (C) Its capillaries lack tight junctions, so it sits outside the blood-brain barrier

Physiology · Emetic center · Step 1

The CTZ in the area postrema is a circumventricular organ whose capillaries lack tight junctions, so it can detect toxins in both blood and CSF. It lies on the floor of the fourth ventricle.

WHY NOT THE OTHERS

(B) The area postrema lies on the floor of the fourth ventricle.

(E) Vagal afferents carry signals from the gut to the NTS, not the other way.

11 (D) Phrenic nerve

Traumatic liver injury · Step 1

Blood or bile under the right hemidiaphragm irritates the diaphragmatic peritoneum, which is supplied by the phrenic nerve (C3–C5). Pain is referred to the C4 dermatome over the shoulder.

12 (A) Non-occlusive mesenteric ischemia

Intestinal ischemia · Step 1

Non-occlusive mesenteric ischemia follows low cardiac output and splanchnic vasoconstriction (shock, vasopressors, digoxin), without a blockage in the vessel. Treatment focuses on improving perfusion; intra-arterial vasodilators may help.

13 (E) The 270° rotation of the midgut around the SMA

Physiology · GI embryology · Step 1

The midgut rotates 270° counterclockwise around the SMA while herniated in weeks 6–10. Malrotation leaves bands and a mesentery that can twist into volvulus, and bilious vomiting, usually the first sign, needs urgent evaluation.

14 (E) Transferrin saturation and serum ferritin

Metabolic liver injury · Step 2

Transferrin saturation (above 45% is abnormal) with ferritin is the initial screen for iron overload. HFE genetic testing follows, which is also offered directly to first-degree relatives. Liver MRI or biopsy is reserved for unclear cases.

15 (A) Enteropathy-associated T-cell lymphoma

Malabsorption · Step 1

Poorly controlled celiac disease raises the risk of enteropathy-associated T-cell lymphoma of the small bowel, as well as small bowel adenocarcinoma. A strict gluten-free diet lowers this risk.

16 (B) Pregnant patients

PUD · Step 1

Misoprostol (a PGE1 analog) increases mucus and bicarbonate and reduces acid. It also causes uterine contractions and can induce abortion, so it

is contraindicated in pregnancy. Its other main side effect is diarrhea.

17 **A** **Metronidazole**

Physiology · Antibacterials · Step 1

Metronidazole replaces amoxicillin in triple therapy for penicillin allergy and causes a disulfiram-like reaction with alcohol, which is avoided during treatment and for 3 days after. It also causes a metallic taste and peripheral neuropathy.

18 **B** **It displaces toxic bile salts and increases bile flow**

Physiology · Liver function · Step 2

Ursodeoxycholic acid is a hydrophilic bile salt that displaces toxic hydrophobic bile salts and increases bile flow, slowing progression in PBC.

19 **E** **Ischemic colitis**

Physiology · Laxatives and antidiarrheals · Step 2

Alosetron, a 5-HT3 antagonist in the gut, is reserved for women with severe IBS-D because of constipation and ischemic colitis, which can be serious. Fluvoxamine, a strong CYP1A2 inhibitor, is contraindicated with it.

WHY NOT THE OTHERS

(B) Pancreatitis is the main risk of eluxadoline.

20 **B** **Fever, right upper quadrant pain, jaundice**

Ascending cholangitis · Step 1

Charcot triad is fever, right upper quadrant pain, and jaundice. Adding hypotension and altered mental status gives Reynolds pentad, which signals septic shock.

WHY NOT THE OTHERS

(A) Painless jaundice with a palpable gallbladder suggests malignancy (Courvoisier law).

(C) RUQ pain, Murphy sign, and fever without jaundice suggest acute cholecystitis.

21 **E** **Enterolithotomy to remove the stone from the bowel**

Biliary fistulas · Step 2

Gallstone ileus is treated by enterolithotomy (opening the bowel to remove the stone), which relieves the obstruction. Cholecystectomy and fistula repair are usually deferred, especially in frail patients, because a combined procedure carries higher risk.

22 **D** **Closer INR checks, because the drug inhibits CYP2C9**

Physiology · Antifungals, antivirals, and antiparasitics · Step 2

Candida esophagitis is treated with systemic fluconazole for 14–21 days, never topically. Fluconazole inhibits CYP2C9, CYP2C19, and CYP3A4, raising warfarin, phenytoin, and statin levels, and it also causes hepatotoxicity and QT prolongation.

WHY NOT THE OTHERS

(E) Fluconazole is given systemically and is well absorbed.

23 **C** **Cholangiocarcinoma**

IBD · Step 1

Primary sclerosing cholangitis is strongly associated with ulcerative colitis (p-ANCA positive) and greatly raises the risk of cholangiocarcinoma. It also raises colorectal cancer risk further, so patients with both need closer colonoscopic surveillance.

24 **B** **Adenocarcinoma of the distal esophagus**

GERD · Step 1

Barrett esophagus is intestinal metaplasia, in which nonciliated columnar epithelium with goblet cells replaces stratified squamous epithelium. It can progress through dysplasia to adenocarcinoma of the distal esophagus, so patients need endoscopic surveillance.

WHY NOT THE OTHERS

(A) Squamous cell carcinoma is linked to smoking and alcohol and typically arises in the upper or middle esophagus.

25 **C** **Retrocecal, against the iliopsoas muscle**

Appendicitis · Step 1

The psoas sign (pain on passive hip extension) means a retrocecal appendix is irritating the iliopsoas. The obturator sign (pain on internal rotation of the flexed hip) suggests a pelvic appendix. Pain migrating from the umbilicus to the RLQ reflects visceral pain becoming parietal as inflammation reaches the peritoneum.

WHY NOT THE OTHERS

(D) A pelvic appendix gives a positive obturator sign.

26 **(D) Intramuscular vitamin B12**

Gastritis · Step 2

Autoimmune gastritis destroys parietal cells, so intrinsic factor is lost and B12 cannot be absorbed normally. Parenteral (IM) B12 replacement bypasses the problem; high-dose oral B12 is an alternative.

WHY NOT THE OTHERS

(A) Folic acid alone can correct the anemia while the neurologic damage from B12 deficiency progresses.

27 **(A) Swansea criteria**

Reading liver function tests · Step 2

Raised transaminases and hypoglycemia in the third trimester suggest acute fatty liver of pregnancy, assessed with the Swansea criteria.

WHY NOT THE OTHERS

(B) The Maddrey score is for alcoholic hepatitis.
(C) Child-Pugh scores cirrhosis.
(D) Rome criteria are for IBS.
(E) Charcot triad is for cholangitis.

28 **(B) Metastatic gastric cancer**

Localizing abdominal pain · Step 1

A Virchow node, like a Sister Mary Joseph nodule or a Krukenberg tumor, points to metastatic gastric cancer.

WHY NOT THE OTHERS

(E) Carcinoid syndrome causes flushing, wheezing, and tricuspid valve disease.

29 **(D) Switching to mesalazine**

Physiology · IBD and immune disease · Step 1

Sulfasalazine causes reversible oligospermia, as well as folate deficiency, rash, agranulocytosis, and hemolysis. Mesalazine is 5-ASA without the sulfapyridine and treats mild to moderate UC for induction and maintenance.

WHY NOT THE OTHERS

(A) Glucocorticoids induce remission but are not used for maintenance.

30 **(B) Blood glucose**

Acute liver failure and cirrhosis · Step 1

Hypoglycemia is one of the "big 4" complications of acute liver failure, alongside encephalopathy, coagulopathy, and infection. The failing liver cannot make glucose or release it from glycogen. A sudden drop in consciousness needs an immediate glucose check.

31 **(E) Bismuth subcitrate**

Physiology · Gastric secretion · Step 2

Bismuth subcitrate coats the ulcer base, kills H. pylori, and increases PG and HCO_3^-.

32 **(E) CT-guided percutaneous drainage plus ABX**

Diverticular diseases · Step 2

Diverticular abscesses larger than about 3–4 cm are drained percutaneously alongside ABX. Surgery is reserved for failed drainage, perforation with peritonitis, or recurrent complicated disease.

33 **(B) Trientine**

Physiology · Antidotes, chelators, and fatty liver · Step 2

Trientine is a copper chelator that causes fewer adverse effects than penicillamine, whose effects include rash, proteinuria, marrow suppression, dysgeusia, drug-induced lupus, and myasthenia. Zinc is slow to act, so it is used for maintenance.

WHY NOT THE OTHERS

(A) Zinc acts slowly, so it is not used alone at first.
(C) Deferoxamine chelates iron.

34 **(E) Dubin-Johnson syndrome**

Reading liver function tests · Step 1

Dubin-Johnson syndrome, from MRP2 defects, raises conjugated bilirubin with a black liver; Rotor syndrome is similar without pigment.

WHY NOT THE OTHERS

(A) Rotor syndrome has no pigment.
(C) Gilbert syndrome raises unconjugated bilirubin.
(D) Crigler-Najjar syndrome raises unconjugated bilirubin.

35 **(A) Ileocecal valve**

Cholecystitis · Step 1

Gallstone ileus: a large stone erodes through a cholecystoenteric fistula (usually cholecystoduodenal) and obstructs the narrowest part of the small bowel, the ileocecal valve. Rigler triad on imaging is pneumobilia, small bowel obstruction, and an ectopic gallstone.

WHY NOT THE OTHERS

(C) A stone obstructing the pylorus or duodenum causes Bouveret syndrome (gastric outlet obstruction), not distal small bowel obstruction.

36 (D) Hepatitis B vaccine plus HBIG within 12 hours of birth

Infectious liver injury | Step 2

Perinatal HBV transmission is highly likely to cause chronic infection (about 90%). Giving both the first vaccine dose and HBIG within 12 hours of birth prevents most infections.

37 (C) ABX therapy

Diarrhea | Step 2

This is hemolytic uremic syndrome from Shiga toxin-producing E. coli (O157:H7). ABX, and antimotility agents such as loperamide, may increase toxin release and the risk of HUS, so they are avoided. Care is supportive: fluids, transfusion, and dialysis as needed.

38 (C) The migrating motor complex

Physiology | Enteric nervous system | Step 1

The MMC, which depends on the ENS, has a quiet phase of about 70 minutes, an intermittent phase of 10–20 minutes, and 5 minutes of regular propulsion, recurring every 90–120 minutes.

39 (D) Goal-directed IV fluids such as Hartmann solution

Acute pancreatitis | Step 2

Third spacing causes hypovolemia in pancreatitis. Early goal-directed IV fluids (Hartmann solution is commonly used), pain control and early feeding are the core of treatment. ABX are only for proven infection, and enteral feeding is preferred over TPN.

WHY NOT THE OTHERS

(B) Prophylactic ABX do not improve outcomes; they are reserved for infected necrosis.

40 (D) Hepatic adenoma

Liver tumors | Step 1

Hepatic adenomas are linked to oral contraceptives and anabolic steroids, and larger ones (over 5 cm) can rupture and bleed. Management includes stopping estrogen and resecting large or bleeding adenomas.

WHY NOT THE OTHERS

(B) Focal nodular hyperplasia has a central stellate scar and rarely bleeds.

41 (E) Upper endoscopy

Esophageal motility disorders | Step 2

Older age, a short history, and marked weight loss suggest pseudoachalasia from a tumor at the gastroesophageal junction. Endoscopy must exclude malignancy before achalasia is diagnosed or treated.

42 (C) Bacillus cereus

Diarrhea | Step 1

B. cereus spores survive cooking. In rice left at room temperature they germinate and form a preformed, heat-stable emetic toxin (cereulide), causing vomiting within 1–6 hours. This is called "reheated rice syndrome".

WHY NOT THE OTHERS

(A) S. aureus also causes rapid vomiting from a preformed toxin, but the classic sources are mayonnaise, potato salad, and custards.

43 (A) Alcoholic hepatitis

Iatrogenic and toxic liver injury | Step 2

Alcoholic hepatitis typically shows an AST:ALT ratio above 2:1 with AST rarely above 300 U/L, raised GGT, and macrocytosis. ALT stays relatively low partly because it needs pyridoxal phosphate (vitamin B6), which is often deficient in heavy drinkers.

WHY NOT THE OTHERS

(D) Viral hepatitis usually causes ALT-predominant rises, often well above 1000 U/L.

(E) MASLD typically has ALT higher than AST.

44 (C) Prophylactic total gastrectomy

Gastric tumors | Step 2

Germline CDH1 mutations cause hereditary diffuse gastric cancer, which is multifocal and grows without a mass, so carriers may have a prophylactic total gastrectomy. Diffuse and signet ring cell cancers need a total rather than subtotal gastrectomy.

45 (C) Pseudohyphae and budding yeast

Esophagitis | Step 1

Candida esophagitis, the most common infectious esophagitis, shows pseudohyphae and budding

yeast. Inhaled steroids, ABX, diabetes, and immunosuppression are risk factors.

WHY NOT THE OTHERS

- (A) Owl's eye inclusions suggest CMV.
- (D) Goblet cells in columnar epithelium describe Barrett esophagus.
- (E) Multinucleated giant cells suggest HSV.

46 **C Paracetamol**

Physiology · Pain and smooth muscle · Step 1

Uncomplicated diverticulitis is treated with paracetamol rather than NSAIDs. Paracetamol inhibits central COX with little anti-inflammatory effect, and its overdose causes hepatotoxicity through NAPQI.

47 **A Clindamycin**

Physiology · When drugs cause disease · Step 1

Clindamycin, cephalosporins, fluoroquinolones, and penicillins are the main causes of C. difficile infection, which oral fidaxomicin or vancomycin then treats.

WHY NOT THE OTHERS

- (B) Oral vancomycin treats C. difficile.
- (E) Fidaxomicin treats C. difficile and spares the colonic flora.

48 **E Hypoglycemia with prolonged INR**

AFLP · Step 2

AFLP causes true liver failure, so hypoglycemia and coagulopathy (raised INR, low fibrinogen) are common. HELLP is defined by hemolysis, raised liver enzymes, and low platelets, usually with hypertension, and rarely causes hypoglycemia. Both are treated with delivery.

49 **D EGD within 24 hours**

GI bleeds and obstruction · Step 2

Upper GI bleeding is investigated by EGD within 24 hours, or 12 hours if variceal, with endoscopic hemostasis and an IV PPI. CTA is used if EGD finds no source or bleeding is brisk.

50 **E MRCP**

Autoimmune liver injury · Step 2

Primary sclerosing cholangitis affects intra- and extrahepatic bile ducts, so it is diagnosed by cholangiography: MRCP shows multifocal strictures and dilations ("beading"). Liver biopsy is reserved for suspected small-duct PSC with a normal MRCP.

WHY NOT THE OTHERS

- (C) A HIDA scan evaluates acute cholecystitis.

Test B answers

Answer key | Explanations

1	A	11	B	21	E	31	C	41	E
2	B	12	B	22	E	32	B	42	A
3	C	13	A	23	E	33	D	43	B
4	D	14	C	24	A	34	C	44	A
5	B	15	B	25	A	35	E	45	C
6	E	16	C	26	E	36	D	46	B
7	C	17	C	27	D	37	B	47	D
8	C	18	D	28	E	38	E	48	D
9	D	19	A	29	C	39	D	49	A
10	B	20	D	30	A	40	E	50	A

1 A Spironolactone

Physiology | Cirrhosis and blood vessels | Step 1

Spironolactone blocks aldosterone receptors in the collecting duct and causes hyperkalemia, gynecomastia, and lower libido. Furosemide, its partner, causes hypokalemia, hypovolemia, and ototoxicity.

2 B Aprepitant inhibits CYP3A4, raising dexamethasone levels

Physiology | Antiemetics | Step 2

NK1 antagonists are metabolized by CYP3A4, which they also inhibit, raising dexamethasone and other CYP3A4 substrate levels. Aprepitant can also weaken warfarin and hormonal contraceptives.

3 C Anti-centromere antibody

Autoimmune liver injury | Step 1

PBC is associated with other autoimmune diseases, including limited systemic sclerosis (CREST: calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, telangiectasia), which is linked to anti-centromere antibodies.

WHY NOT THE OTHERS

(D) Anti-Scl-70 is associated with diffuse systemic sclerosis.

4 D Skin, then liver, then kidney

Physiology | Liver function | Step 1

The skin makes cholecalciferol, the liver forms 25-OH cholecalciferol, and the kidney forms active calcitriol.

5 B Serum triglycerides of 2400 mg/dL

Acute pancreatitis | Step 1

Hypertriglyceridemia (usually over 1000 mg/dL) is a cause of acute pancreatitis (the H in I GET SMASHED). Lipase breaks down the triglycerides into toxic free fatty acids that injure the pancreas. Milky (lipemic) serum is a clue. Treatment includes insulin infusion to lower triglycerides.

WHY NOT THE OTHERS

(D) Hypercalcemia (not low calcium) causes pancreatitis; low calcium is a complication of it.

6 E Sodium restriction and spironolactone, with furosemide added as needed

Acute liver failure and cirrhosis | Step 2

First-line ascites treatment is dietary sodium restriction plus spironolactone (blocking secondary hyperaldosteronism), often with furosemide. Large-volume paracentesis with albumin, TIPS, or transplant are used for tense or refractory ascites.

7 **C** **Hypochloremic, hypokalemic metabolic alkalosis**

Physiology · GI embryology · Step 2

Pyloric stenosis causes projectile, nonbilious vomiting at 3–6 weeks with an olive in 60–80%, and the classic electrolyte picture is hypochloremic, hypokalemic metabolic alkalosis. Treatment is pyloromyotomy.

8 **C** **Therapeutic phlebotomy**

Metabolic liver injury · Step 2

Regular phlebotomy is first-line: each unit of blood removes iron, and it is continued until ferritin is in the target range. Chelation is reserved for patients who cannot tolerate phlebotomy, such as those with anemia or heart failure.

WHY NOT THE OTHERS

(D) D-penicillamine chelates copper and is used for Wilson disease.

9 **D** **Use blood products without IgA**

Physiology · Gut immunity · Step 2

Selective IgA deficiency, about 1 in 700, causes sinopulmonary infections and giardiasis. Anti-IgA antibodies can cause anaphylaxis after blood products containing IgA.

WHY NOT THE OTHERS

(A) IV immunoglobulin contains IgA and could itself trigger a reaction.

10 **B** **CT angiography of the abdomen**

Intestinal ischemia · Step 2

Pain out of proportion to examination in a patient with an embolic source suggests acute mesenteric ischemia. CT angiography is the diagnostic test of choice. Treatment is urgent revascularization (endovascular or surgical) with anticoagulation, plus resection of necrotic bowel.

11 **B** **Squamous cell carcinoma of the esophagus**

Esophageal motility disorders · Step 1

Long-standing achalasia raises the risk of esophageal squamous cell carcinoma, probably from chronic irritation by stagnant food. Aspiration pneumonia is the other key complication.

12 **B** **Diverticulosis**

Diverticular diseases · Step 1

Diverticulosis is the most common cause of major lower GI bleeding in older adults. It is typically painless, sudden, and large in volume because a vasa recta artery ruptures. Most bleeds stop on their own; colonoscopy finds and treats the source.

WHY NOT THE OTHERS

(A) An anal fissure causes painful, small-volume bleeding.

(C) Ischemic colitis causes pain with bleeding.

13 **A** **Graded-compression ultrasound, followed by MRI if inconclusive**

Appendicitis · Step 2

In pregnancy, ultrasound is used first to avoid radiation, then MRI without gadolinium if ultrasound is inconclusive. The appendix may be displaced upward by the uterus, so pain can be atypical. Delaying treatment risks perforation, which threatens the fetus.

14 **C** **Upper endoscopy with biopsy**

GERD · Step 2

Dysphagia, weight loss, and iron-deficiency anemia are alarm features that require endoscopy with biopsy to exclude malignancy, rather than an empiric PPI trial.

15 **B** **Hepatopancreatic ampulla (of Vater)**

Choledocholithiasis · Step 1

Gallstone pancreatitis occurs when a stone blocks the ampulla, where the common bile duct and pancreatic duct meet, causing pancreatic enzyme activation. A cholestatic LFT pattern with high lipase and gallstones on ultrasound suggests a gallstone cause; it is the most common cause of acute pancreatitis.

WHY NOT THE OTHERS

(C) A cystic duct stone causes biliary colic or cholecystitis with normal LFTs and lipase.

16 **C** **Hypovolemia leading to gastric mucosal ischemia**

Gastritis · Step 1

Curling ulcers follow severe burns. Massive fluid loss causes hypovolemia and gastric mucosal ischemia, which leads to erosions. Burn = Curling ("cooking"); brain = Cushing.

WHY NOT THE OTHERS

(D) Increased vagal stimulation explains Cushing ulcers after brain injury or raised ICP.

17 (C) Bile salts entering the colon directly

Biliary fistulas · Step 1

A cholecystocolic fistula lets bile bypass the small intestine, where bile salts are normally reabsorbed. Bile salts reaching the colon stimulate water secretion, causing bile acid diarrhea; bile also fails to reach the small bowel, so fat absorption suffers.

18 (D) Urgent ERCP for biliary drainage

Ascending cholangitis · Step 2

Charcot triad (fever, RUQ pain, jaundice) plus hypotension and confusion is Reynolds pentad: severe ascending cholangitis with septic shock. ABX alone are not enough; the infected, obstructed bile must be drained urgently, usually by ERCP.

19 (A) Empiric oral fluconazole

Esophagitis · Step 2

Candida is the most common cause of infectious esophagitis. In an immunocompromised patient with odynophagia and oral thrush, empiric systemic fluconazole is appropriate. Endoscopy is reserved for patients who do not improve within about 72 hours.

WHY NOT THE OTHERS

(B) Acyclovir treats HSV esophagitis (small "punched-out" ulcers); oral thrush points to Candida.

(C) Ganciclovir treats CMV esophagitis (large, shallow linear ulcers).

(E) EGD is indicated only if empiric antifungal therapy fails.

20 (D) The mid-esophagus

Physiology · When drugs cause disease · Step 1

NSAIDs, KCl, doxycycline, and bisphosphonates cause pill esophagitis by lodging in the mid-esophagus, especially when taken lying down with little water.

21 (E) Stopping cannabis

Physiology · Emetic center · Step 2

Years of cannabis use, cyclical vomiting relieved by hot showers, and wellness between episodes suggest cannabinoid hyperemesis syndrome. It resolves after stopping cannabis.

WHY NOT THE OTHERS

(A) Cyclic vomiting syndrome is linked with migraine, but the hot showers and daily cannabis point elsewhere.

(C) Scopolamine prevents motion sickness.

22 (E) Reassurance; no biopsy or treatment needed

Liver tumors · Step 2

Peripheral nodular enhancement with centripetal fill-in is typical of a hemangioma, the most common benign liver tumor. Asymptomatic hemangiomas need no treatment, and biopsy is avoided because of bleeding risk.

23 (E) Emergency surgical repair

PUD · Step 2

Free air with peritonitis means a perforated ulcer, which is a surgical emergency. Resuscitate with IV fluids, IV PPI, and broad-spectrum ABX, then proceed to surgery.

WHY NOT THE OTHERS

(A) Barium leaking into the peritoneum causes severe peritonitis; water-soluble contrast is used if contrast is needed at all.

(B) Endoscopy insufflates air and can worsen a known perforation.

24 (A) E-cadherin

Gastric tumors · Step 1

Diffuse-type gastric adenocarcinoma lacks cell adhesion from CDH1 (E-cadherin) loss, so signet ring cells infiltrate the wall without a mass, producing linitis plastica, the leather bottle stomach. It has no precursor lesion and affects men and women equally.

WHY NOT THE OTHERS

(C) KIT mutations drive GISTs.

(E) HER2 is amplified in some gastric cancers and guides targeted treatment; it does not explain loss of adhesion.

25 (A) Clarithromycin

Physiology · Antibacterials · Step 2

Clarithromycin is a strong CYP3A4 inhibitor that raises levels of statins (myopathy), calcineurin inhibitors, carbamazepine, and warfarin, so it is contraindicated with simvastatin and lovastatin.

Resistance to it is also the main reason triple therapy fails.

26 **E** **Microvesicular steatosis**

AFLP | Step 1

AFLP shows microvesicular steatosis: tiny fat droplets fill hepatocytes without displacing the nucleus, reflecting mitochondrial failure of fatty acid oxidation.

WHY NOT THE OTHERS

- (A) Macrovesicular steatosis with Mallory-Denk bodies suggests alcoholic liver disease.
- (D) Periportal fibrin deposits and hemorrhage suggest HELLP or pre-eclampsia.

27 **D** **Bedside ultrasound**

Localizing abdominal pain | Step 2

Pain with hypotension and a pulsatile mass suggests a ruptured AAA. Bedside ultrasound is used if the patient is unstable, and CT if stable.

WHY NOT THE OTHERS

- (A) Shock with a pulsatile mass should not be attributed to a stone.
- (B) CT is for a stable patient; he is too unstable to leave the resuscitation area.
- (E) Lipase is for pancreatitis.

28 **E** **IV third-generation cephalosporin plus IV albumin**

Acute liver failure and cirrhosis | Step 2

An ascitic neutrophil count of 250/μL or more diagnoses spontaneous bacterial peritonitis. Treat with a third-generation cephalosporin (e.g. ceftriaxone or cefotaxime). IV albumin reduces the risk of hepatorenal syndrome. Long-term ABX prophylaxis follows.

WHY NOT THE OTHERS

- (C) SBP has no surgical source; laparotomy is for secondary peritonitis such as a perforation.

29 **C** **CT abdomen with IV contrast**

Traumatic liver injury | Step 2

A hemodynamically stable patient with suspected blunt abdominal trauma gets a contrast CT to identify and grade solid-organ injury. Most stable liver injuries are managed without surgery, with angioembolization if there is active bleeding.

WHY NOT THE OTHERS

- (E) Emergency laparotomy is for unstable patients with a positive FAST or peritonitis.

30 **A** **Stomach**

Gastric tumors | Step 1

A Krukenberg tumor is a signet ring cell metastasis to the ovaries, often bilateral, most often from gastric adenocarcinoma. Other signs of spread include the Virchow, Sister Mary Joseph, and Irish nodes and a Blumer shelf.

31 **C** **Lactase replacement**

Physiology | Digestion and absorption | Step 2

Lactase deficiency leaves lactose undigested, causing osmotic diarrhea and gas from bacterial digestion. Treatment is lactase replacement or a lactose-free diet.

32 **B** **Ultrasound**

Localizing abdominal pain | Step 2

For suspected appendicitis, CT AP with contrast is used in adults, but ultrasound comes first in children and pregnancy.

WHY NOT THE OTHERS

- (C) CT is avoided as the first test in pregnancy.
- (D) An upright CXR is for suspected perforation.
- (E) Colonoscopy has no role in suspected appendicitis.

33 **D** **IV ABX and laparoscopic cholecystectomy during this admission**

Cholecystitis | Step 2

This is acute calculous cholecystitis: prolonged RUQ pain, fever, Murphy sign, leukocytosis, typical ultrasound findings, and normal LFTs. Treat with supportive care and ABX, followed by early laparoscopic cholecystectomy, ideally within 72 hours.

WHY NOT THE OTHERS

- (B) ERCP is for common bile duct stones; normal bilirubin and ALP make this unlikely.
- (E) Cholecystostomy is reserved for patients too unwell for surgery or with acalculous cholecystitis.

34 **C** **It increases cholesterol secretion into bile**

Cholelithiasis | Step 1

Estrogen increases hepatic cholesterol uptake and synthesis (upregulating HMG-CoA reductase), so more cholesterol enters bile. Progesterone in pregnancy also slows gallbladder emptying. Both help explain the "female" and "fertile" Fs.

35 **E** **IgA anti-tTG antibodies with total serum IgA**

Malabsorption · Step 2

Screen for celiac disease with IgA anti-tTG while the patient is still eating gluten. Check total IgA because IgA deficiency (more common in celiac disease) gives false-negative results. Confirm with EGD and duodenal biopsy.

WHY NOT THE OTHERS

(B) Biopsy confirms the diagnosis but follows a positive serology.

36 **D** **Replenishing glutathione to neutralize NAPQI**

Iatrogenic and toxic liver injury · Step 1

In overdose, glucuronidation and sulfation are saturated, so more paracetamol is converted by CYP2E1 to the toxic metabolite NAPQI. Once glutathione is used up, NAPQI damages hepatocytes (zone 3 necrosis). N-acetylcysteine restores glutathione and is most effective within 8 hours.

WHY NOT THE OTHERS

(C) Binding the drug in the gut describes activated charcoal, which helps only within 1–2 hours.

37 **B** **Blood proteins digested and absorbed from the upper gut**

GI bleeds and obstruction · Step 1

In upper GI bleeding, blood is digested and absorbed, so a BUN:creatinine ratio over 30 points to an upper source. Melena, black tarry stool, also points upward, and peptic ulcer disease from NSAIDs or H. pylori is the most common cause.

WHY NOT THE OTHERS

(D) Liver failure lowers urea, which would lower the ratio.

38 **E** **It replenishes glutathione to detoxify NAPQI**

Physiology · Antidotes, chelators, and fatty liver · Step 1

NAC replenishes glutathione, which detoxifies NAPQI, and is nearly 100% effective within 8 hours. IV NAC can cause rate-related anaphylactoid reactions; activated charcoal helps only within 1–2 hours.

WHY NOT THE OTHERS

(C) That is activated charcoal, which helps only within 1–2 hours.

39 **D** **Hepatitis E virus**

Infectious liver injury · Step 1

HEV spreads by the fecal-oral route (contaminated water) and is usually self-limited, but in pregnancy, particularly the third trimester, it can cause fulminant hepatitis with high mortality. It is a naked +ssRNA hepevirus.

WHY NOT THE OTHERS

(C) HAV is also fecal-oral, but it is not associated with fulminant hepatitis in pregnancy.

40 **E** **Untreated B12 deficiency progresses despite the better anemia**

Physiology · Replacement · Step 2

Folic acid corrects the anemia of B12 deficiency while the neuropathy progresses, so untreated B12 deficiency is a contraindication to giving it alone. Folic acid also lowers phenytoin levels.

WHY NOT THE OTHERS

(C) Folic acid lowers phenytoin levels.

41 **E** **Sucralfate needs acid to form its paste**

Physiology · Acid and the mucosa · Step 2

Sucralfate polymerizes at pH under 4 into a paste that binds the ulcer base, so it works poorly with PPIs, H2R antagonists, and antacids. It also binds digoxin, levothyroxine, fluoroquinolones, furosemide, and warfarin.

WHY NOT THE OTHERS

(D) Sucralfate is barely absorbed; omeprazole is the CYP2C19 inhibitor.

42 **A** **Allopurinol blocks xanthine oxidase, which breaks down 6-MP**

Physiology · IBD and immune disease · Step 2

Azathioprine becomes 6-MP, which xanthine oxidase breaks down, so allopurinol and febuxostat sharply raise thiopurine levels and marrow toxicity. TPMT activity is tested before starting thiopurines.

43 **B** **Cardiac monitoring with atropine at hand**

Physiology · Motility · Step 2

Neostigmine blocks AChE, causing bradycardia and asystole, bronchospasm, salivation, and cramps, so it is given with cardiac monitoring and atropine at hand. It is contraindicated in mechanical obstruction and peritonitis.

44 **A** **Anticoagulation and stopping the oral contraceptive**

Vascular liver injury · Step 2

All patients with Budd-Chiari syndrome are anticoagulated, and the underlying cause is treated (here, stopping estrogen). Treatment then steps up if needed: angioplasty or stenting for short venous narrowing, then TIPS, then liver transplant for fulminant failure or cirrhosis.

45 **C** **Shigella**

Diarrhea · Step 1

Shigella has a very low infectious dose (as few as 10 organisms), so it spreads easily person to person. It invades colonic mucosa, causing dysentery, and is non-motile and does not produce H_2S.

WHY NOT THE OTHERS

(E) Salmonella is motile, produces H_2S, and usually comes from poultry or eggs.

46 **B** **Albumin has a half-life of 14–21 days**

Reading liver function tests · Step 1

Albumin, made only by the liver, has a half-life of 14–21 days, so it falls in chronic, not acute, liver disease. Clotting factors have short half-lives, so the INR rises quickly when synthesis fails.

WHY NOT THE OTHERS

(A) Inflammation lowers albumin.

(E) Albumin is made only by the liver.

47 **D** **Inhibition through thoracolumbar sympathetic fibers**

Physiology · Gastric secretion · Step 1

Sympathetic fibers from the thoracolumbar outflow inhibit stomach activity, opposing the stimulation from efferent CN X.

48 **D** **Interferon-gamma release assay (or tuberculin skin test)**

IBD · Step 2

Infliximab is a TNF-α inhibitor. TNF-α is essential for maintaining granulomas, so blocking it can reactivate latent tuberculosis. Screen with an IGRA or tuberculin skin test (and for hepatitis B) before starting.

WHY NOT THE OTHERS

(B) TPMT activity is checked before azathioprine or 6-MP, not before anti-TNF therapy.

49 **A** **Dry mouth and constipation**

IBS · Step 1

TCAs block muscarinic receptors, causing dry mouth, constipation, urinary retention, and blurred vision. They also cause sedation (via H_1) and orthostasis (via α_1). The constipating effect can be useful in IBS-D, and TCAs reduce visceral hypersensitivity.

50 **A** **Pancreaticoduodenectomy (Whipple procedure)**

Biliary malignancies · Step 2

A resectable tumor in the pancreatic head or periampullary region (the "double duct sign" shows both ducts blocked) is treated with pancreaticoduodenectomy (Whipple procedure), usually with chemotherapy. Stenting relieves jaundice but does not treat the cancer.

WHY NOT THE OTHERS

(D) Distal pancreatectomy is used for tumors of the pancreatic body and tail.

Test C answers

Answer key | Explanations

1 A	11 A	21 E	31 D	41 C
2 A	12 C	22 C	32 D	42 B
3 B	13 A	23 B	33 C	43 E
4 A	14 E	24 C	34 E	44 B
5 B	15 A	25 A	35 D	45 B
6 D	16 E	26 B	36 C	46 A
7 B	17 B	27 A	37 E	47 E
8 E	18 B	28 D	38 E	48 D
9 E	19 D	29 A	39 C	49 C
10 D	20 C	30 C	40 D	50 D

1 A Noncaseating granulomas with transmural inflammation

IBD | Step 1

Crohn disease shows skip lesions, transmural inflammation (causing fistulas and strictures), noncaseating granulomas, and Th1-mediated inflammation. It most often involves the terminal ileum and spares the rectum. Smoking worsens Crohn disease.

WHY NOT THE OTHERS

- (C) Caseating granulomas suggest intestinal tuberculosis.
- (D) Inflammation limited to the mucosa and submucosa, with crypt abscesses, is typical of ulcerative colitis.

2 A HCV RNA by PCR

Infectious liver injury | Step 2

A positive anti-HCV antibody shows exposure but not whether the infection is current; the antibody stays positive for life, even after cure. HCV RNA confirms active infection. If positive, assess fibrosis (e.g. APRI, FIB-4, or elastography) before treatment.

WHY NOT THE OTHERS

- (D) Anti-HCV antibodies are not protective; reinfection is possible.

3 B A necrotic Peyer patch

Physiology | Gut immunity | Step 1

In typhoid, Salmonella targets M cells, and necrosis of Peyer patches can perforate the distal ileum, most often in the third week of fever.

4 A No further test is needed

Acute pancreatitis | Step 2

Acute pancreatitis is diagnosed with 2 of 3 criteria: typical epigastric pain, lipase at least 3 times normal, and typical imaging findings. He already meets two. Early CT is not needed and can miss necrosis; it is reserved for an unclear diagnosis or for patients who fail to improve after 48–72 hours. A RUQ ultrasound is still done to look for gallstones.

WHY NOT THE OTHERS

- (B) CT within the first 48–72 hours adds little when the diagnosis is already clear and can underestimate necrosis.

5 B Antacids

Physiology | Gastric secretion | Step 2

Antacids neutralize HCl in the lumen, increasing gastric pH and giving fast, short-acting relief.

6 **D** **Vinyl chloride**

Liver tumors · Step 1

Hepatic angiosarcoma, a malignant vascular tumor, is linked to vinyl chloride (used to make PVC plastics), arsenic and Thorotrast.

WHY NOT THE OTHERS

(A) Aflatoxin is linked to hepatocellular carcinoma.

(B) Aniline dyes are linked to bladder cancer.

7 **B** **Bacterial fermentation acidifies the colon, trapping NH4+**

Physiology · Laxatives and antidiarrheals · Step 1

Colonic bacteria ferment lactulose to acids, trapping ammonia as NH4+, and its osmotic effect speeds transit. It is titrated to 2–3 soft stools a day and is contraindicated in galactosemia.

WHY NOT THE OTHERS

(E) That describes rifaximin, which is added if lactulose alone fails.

8 **E** **QT prolongation**

Physiology · Antiemetics · Step 1

5-HT3 antagonists cause headache, constipation, and QT prolongation, and can cause serotonin syndrome with other serotonergic drugs. They are contraindicated with apomorphine and in congenital long QT syndrome.

WHY NOT THE OTHERS

(B) Ischemic colitis is the main risk of alosetron, a 5-HT3 antagonist used in IBS-D.

(D) Tardive dyskinesia follows D2 antagonists such as metoclopramide.

9 **E** **Prednisolone**

Iatrogenic and toxic liver injury · Step 2

Severe alcoholic hepatitis (Maddrey discriminant function of 32 or more) is treated with prednisolone if there is no infection or active bleeding. Abstinence, nutrition, and thiamine are essential for all patients.

10 **D** **Emergency laparotomy**

Traumatic liver injury · Step 2

A hemodynamically unstable trauma patient with a positive FAST needs emergency laparotomy to control bleeding. Unstable patients should not be sent to CT.

11 **A** **Good hydration during treatment**

Physiology · Antifungals, antivirals, and antiparasitics · Step 1

Acyclovir is renally cleared and causes crystal nephropathy, which hydration prevents, and neurotoxicity in renal failure, so its dose is lowered in renal impairment. Its nephrotoxicity adds to that of other kidney-toxic drugs.

WHY NOT THE OTHERS

(C) Oral bioavailability is low, and IV treatment is needed here.

(E) NSAIDs add to the nephrotoxicity.

12 **C** **Conjugated bilirubin reaching the urine**

Physiology · Liver function · Step 2

Only conjugated bilirubin reaches the urine, so dark urine points to conjugated hyperbilirubinemia. Stool is pale if bile is obstructed.

13 **A** **Escherichia coli**

Ascending cholangitis · Step 1

Ascending cholangitis is caused by gut organisms climbing into obstructed bile, most often E. coli, followed by Klebsiella, Enterococcus, and anaerobes. Empiric ABX (e.g. piperacillin-tazobactam) cover these.

14 **E** **The nucleus tractus solitarius**

Physiology · Emetic center · Step 1

The CTZ, vagal afferents, the vestibular system, and higher centers converge on the NTS, which begins the final common pathway and drives the central pattern generator in the medullary reticular formation.

15 **A** **Pregabalin**

Physiology · Pain and smooth muscle · Step 2

Pain in chronic pancreatitis is treated first with paracetamol, NSAIDs, or pregabalin, and opioids only if these fail. Pregabalin binds the α2δ-1 subunit of calcium channels and is renally cleared unchanged; pancreatic enzymes treat steatorrhea but not pain.

WHY NOT THE OTHERS

(C) Enzymes treat steatorrhea; they do not relieve pain.

16 E Loperamide

IBS · Step 2

Loperamide, a peripherally acting μ-opioid agonist, slows gut transit and treats diarrhea in IBS-D. Laxatives are used for constipation-predominant IBS.

17 B Blocking H2 receptors, lowering cAMP in parietal cells

GERD · Step 1

H2-receptor antagonists competitively block histamine H2 receptors on parietal cells, reducing cAMP and acid secretion. They are less potent than PPIs and suit milder symptoms.

18 B Rovsing sign

Appendicitis · Step 1

Rovsing sign is RLQ pain on palpation of the LLQ, from displacing bowel contents and stretching the inflamed peritoneum. It supports the diagnosis of appendicitis.

WHY NOT THE OTHERS

- (C) Cullen sign is periumbilical bruising from retroperitoneal bleeding.
- (E) Murphy sign is inspiratory arrest on RUQ palpation in cholecystitis.

19 D Corticosteroids

Autoimmune liver injury · Step 2

IgG4-related sclerosing cholangitis often occurs with autoimmune pancreatitis and can mimic PSC or cancer. Unlike PSC, it responds well to corticosteroids.

20 C Portal vein

Vascular liver injury · Step 1

The liver has a dual blood supply: the portal vein provides about 75% of hepatic blood flow, and the hepatic artery about 25% (but roughly half the oxygen). This dual supply makes hepatic infarction uncommon.

21 E Prompt delivery

AFLP · Step 2

Hypoglycemia, coagulopathy, and encephalopathy in the third trimester point to acute fatty liver of pregnancy. The only cure is prompt delivery, with supportive care (glucose, blood products, ICU). The newborn should be tested for LCHAD deficiency.

WHY NOT THE OTHERS

- (C) Ursodeoxycholic acid treats intrahepatic cholestasis of pregnancy, which causes itching but not liver failure.

22 C Rectal suction biopsy

Physiology · Enteric nervous system · Step 2

Hirschsprung disease follows failed migration of neural crest cells, leaving the rectosigmoid without ganglion cells in about 80%. It is diagnosed by rectal suction biopsy, with calretinin staining.

WHY NOT THE OTHERS

- (B) A sweat test investigates meconium ileus from cystic fibrosis.

23 B Staging laparoscopy with peritoneal washings

Gastric tumors · Step 2

Staging laparoscopy with peritoneal washings finds peritoneal disease in up to 40% of patients that imaging missed, which changes treatment. Deeper or node-positive tumors are then treated with perioperative chemotherapy and gastrectomy with D2 lymphadenectomy.

WHY NOT THE OTHERS

- (A) Endoscopic resection suits tumors confined to the mucosa or superficial submucosa.
- (D) CEA and CA 19-9 add little to the diagnosis of gastric cancer.

24 C Urgent surgery

GI bleeds and obstruction · Step 2

Constant pain, fever, tachycardia, leukocytosis, acidosis, or peritonitis signal strangulation, which needs surgery, as do ischemia, perforation, and complete obstruction. Partial obstruction often settles without it.

25 A Gallbladder carcinoma

Biliary malignancies · Step 1

A porcelain gallbladder (calcified wall from chronic inflammation) is associated with gallbladder carcinoma, as are large gallstones, polyps over 1 cm, and PSC. Cholecystectomy is often recommended.

26 **B** **CTR1, then ATP7A**

Physiology | Digestion and absorption | Step 1

Copper enters the enterocyte through CTR1 and reaches the blood through ATP7A.

27 **A** **PPI + clarithromycin + metronidazole**

PUD | Step 2

In penicillin allergy, metronidazole replaces amoxicillin in triple therapy, giving PPI + clarithromycin + metronidazole.

WHY NOT THE OTHERS

(E) Amoxicillin is a penicillin and is contraindicated after anaphylaxis.

28 **D** **Bouveret syndrome**

Biliary fistulas | Step 2

Bouveret syndrome is gastric outlet obstruction caused by a gallstone that has passed through a cholecystoduodenal (or cholecystogastric) fistula and lodged in the pylorus or duodenum. Endoscopic removal is tried first; surgery if it fails.

WHY NOT THE OTHERS

(E) Mirizzi syndrome is extrinsic compression of the common hepatic duct by a stone in the cystic duct, causing jaundice.

29 **A** **Hepatocyte copper excretion into bile**

Metabolic liver injury | Step 1

Wilson disease is caused by ATP7B mutations (autosomal recessive). ATP7B moves copper into bile and onto ceruloplasmin. Without it, copper builds up in the liver and then deposits in the basal ganglia (movement and psychiatric symptoms) and cornea (Kayser-Fleischer rings).

30 **C** **Constricting splanchnic vessels through V1 receptors**

Physiology | Cirrhosis and blood vessels | Step 2

Terlipressin, a vasopressin analog, constricts splanchnic vessels through V1 receptors, redirecting blood to the kidneys. It causes ischemia of the gut, fingers, or heart and is avoided with hypoxia or volume overload.

31 **D** **Midgut**

Physiology | GI embryology | Step 1

The celiac artery supplies the foregut, the SMA the midgut, from the mid-duodenum to the proximal two-thirds of the transverse colon, and the IMA the hindgut.

32 **D** **Eosinophilic esophagitis**

Esophagitis | Step 1

Eosinophilic esophagitis occurs in atopic patients (asthma, allergic rhinitis, atopic dermatitis) and is driven by food allergens. Classic findings are food impaction, "trachealized" rings and furrows, and at least 15 eosinophils/hpf. Management includes eliminating food allergens, PPIs, and swallowed topical steroids.

WHY NOT THE OTHERS

(B) A Schatzki ring is a single distal mucosal ring and does not explain dense eosinophilic infiltration.

(E) GERD can cause some eosinophils, but rings, furrows, atopy, and food impaction point to EoE.

33 **C** **Increased estrogen from reduced hepatic clearance**

Acute liver failure and cirrhosis | Step 1

The cirrhotic liver clears estrogen poorly, and more androgen is converted to estrogen peripherally. The excess estrogen causes gynecomastia, testicular atrophy, palmar erythema, and spider angiomas.

34 **E** **A small fraction is absorbed passively without IF**

Physiology | Replacement | Step 1

In autoimmune gastritis, B12 is given IM 1000 µg daily or on alternate days for 1–2 weeks, then IM or high-dose oral for life, because high oral doses work by passive absorption without IF. PPIs, H2R antagonists, and metformin lower absorption of dietary B12.

35 **D** **MRCP or endoscopic ultrasound**

Reading liver function tests | Step 2

A cholestatic pattern with RUQ pain and a dilated CBD suggests choledocholithiasis, investigated with MRCP or endoscopic ultrasound, or ERCP if likelihood is high.

WHY NOT THE OTHERS

(B) AMA is for PBC, which has normal ducts.

(E) HIDA is for cholecystitis that is unclear on ultrasound.

36 **C** **IV ABX and percutaneous cholecystostomy**

Cholecystitis · Step 2

Acalculous cholecystitis occurs in critically ill patients from gallbladder stasis and ischemia, and carries a high risk of gangrene and perforation. Unstable patients are treated with ABX and percutaneous drainage (cholecystostomy).

37 **E** **MRCP or endoscopic ultrasound**

Choledocholithiasis · Step 2

With an intermediate likelihood of a CBD stone (abnormal LFTs or a dilated duct, but no stone seen and no cholangitis), less invasive imaging with MRCP or EUS comes first. ERCP is reserved for high likelihood or a confirmed stone, because it carries risks such as pancreatitis.

38 **E** **Gastric adenocarcinoma**

PUD · Step 1

CagA-positive H. pylori strains cause more severe inflammation and carry a higher risk of peptic ulcers and gastric adenocarcinoma. VacA is another cytotoxin associated with more severe disease.

39 **C** **Hypomagnesemia**

Physiology · Acid and the mucosa · Step 1

Long-term PPI use causes hypomagnesemia, along with B12 and iron deficiency, C. difficile infection, pneumonia, and fractures.

WHY NOT THE OTHERS

(B) Hypermagnesemia follows magnesium antacids or laxatives in renal failure.

(D) Calcium antacids, not PPIs, cause hypercalcemia.

40 **D** **HLA-DQ2 and DQ8**

Malabsorption · Step 1

Celiac disease is associated with HLA-DQ2 and DQ8. Gliadin (from wheat, rye, and barley) is deamidated by tissue transglutaminase and presented on these HLA molecules, driving a T-cell response that damages the mucosa. The rash is dermatitis herpetiformis. Iron deficiency reflects duodenal involvement.

WHY NOT THE OTHERS

(B) HLA-B27 is linked to the seronegative spondyloarthropathies.

(C) HLA-DR3/DR4 is classically linked to type 1 diabetes.

41 **C** **Falling IgA anti-tTG levels**

Malabsorption · Step 2

Treatment is a lifelong gluten-free diet (avoiding wheat, rye, and barley). Celiac antibodies such as IgA anti-tTG fall with good adherence, and persistently raised levels suggest ongoing gluten exposure.

WHY NOT THE OTHERS

(A) HLA type does not change; HLA-DQ2/DQ8 testing helps rule celiac disease out, not monitor it.

42 **B** **It is an HDV superinfection and is likely to become chronic**

Infectious liver injury · Step 1

HDV needs HBsAg to form its envelope, so it only infects people with HBV. A negative IgM anti-HBc shows the HBV is chronic, so this is a superinfection, which is usually more severe and becomes chronic. In coinfection, both are acquired together and IgM anti-HBc is positive.

WHY NOT THE OTHERS

(A) Coinfection would show a positive IgM anti-HBc, because the HBV infection would be acute.

(C) There is no hepatitis C vaccine; HDV is prevented by the hepatitis B vaccine.

43 **E** **No treatment; return if symptoms develop**

Cholelithiasis · Step 2

Most gallstones never cause symptoms, and the risk of surgery outweighs the benefit in asymptomatic patients. Cholecystectomy is offered once symptoms (such as biliary colic) or complications occur.

44 **B** **Oral rehydration, with loperamide if needed**

Diarrhea · Step 2

Mild traveler's diarrhea, most often from enterotoxigenic E. coli, is self-limited. Oral rehydration is the mainstay, with loperamide for symptom control if there is no fever or blood. ABX are reserved for moderate to severe cases.

45 **B** **NS3/4A protease inhibitors**

Physiology · Hepatitis and bile salts · Step 2

NS3/4A protease inhibitors such as glecaprevir and grazoprevir are avoided in decompensated cirrhosis (Child-Pugh B or C). They are also strong

CYP3A4 inhibitors that raise calcineurin inhibitor levels.

46 **A** **Esophageal manometry**

Esophageal motility disorders · Step 2

Esophageal manometry is the gold standard. In achalasia it shows incomplete LES relaxation, raised LES pressure, and absent distal peristalsis. Endoscopy is done first to exclude malignancy at the GE junction (pseudoachalasia).

47 **E** **Prophylactic PPI or H2 blocker**

Gastritis · Step 2

Critically ill patients, especially those on mechanical ventilation, with coagulopathy, burns, or head injury (Cushing ulcers), are at risk of stress ulcer bleeding. Acid suppression with a PPI or H2 blocker is used as prophylaxis.

48 **D** **Referred pain from nerves sharing a spinal level**

Localizing abdominal pain · Step 1

Blood from a ruptured spleen irritates the diaphragm, and the shoulder pain is referred because visceral and somatic nerves share a spinal level. Left shoulder pain with splenic rupture is the Kehr sign. He needs resuscitation and imaging.

WHY NOT THE OTHERS

- (B) He did not injure the shoulder, and the timing follows the abdominal injury.
- (C) Somatoparietal pain is felt at the site of the inflamed peritoneum.
- (E) Visceral pain is dull and poorly localized to the abdomen, not felt at the shoulder.

49 **C** **Splenic flexure**

Intestinal ischemia · Step 1

Ischemic colitis from hypoperfusion affects watershed areas. The splenic flexure (between the SMA and IMA territories) and the rectosigmoid junction are most vulnerable. The distal rectum is spared because it has a dual blood supply from the IMA and the iliac branches.

WHY NOT THE OTHERS

- (E) The distal rectum has a dual blood supply and is usually spared.

50 **D** **Colovesical fistula**

Diverticular diseases · Step 2

Pneumaturia, fecaluria, and recurrent polymicrobial UTIs suggest a colovesical (enterovesical) fistula, most often from diverticulitis. CT is the usual first test (air in the bladder), and treatment is surgical.